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Energy Futures Lab

Climate Resilience and Electricity Systems: Visualising Future Climate Impacts in the UK

Author:
Katerina Foka Sandoval

Supervisor:
Dr. Malte Jansen

Co-Supervisor:
Dr. Fei Teng

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Abstract

Electricity systems carry 90% of humanity's final energy use; they are essential for the smooth operations of crucial wide-scale services including food production and retail, transportation, health and telecommunications. Their failure can have severe and widespread impact as was illustrated in 2021 during the Texas Power Crisis. They are, nevertheless, regularly exposed to weather hazards and will be increasingly threatened by the effects of climate change. In order to provide an accurate landscape of potential weather-related impacts on electricity-grid components, a literature research was carried out, based on scientific and industry reports. The references and the associated impacts were categorised according to the analysed weather variables and affected grid components. The weather-pressures which are reported on the most are ambient-temperature increase, heat waves and reduced water availability.

A lack of nation-wide adaptation roadmaps was identified for the UK's electricity sector. Drawing on the conducted review, the hypothesis is substantiated that this could be the result of a lack of intuitive content laying out the potential impacts of weather pressures on the UK's electricity grid components. In response, this thesis work aimed to deliver an adaptable program that can be used to quantify impacts on custom grid-component and weather-variable sets, as well as an associated visualisation tool.

An impact assessment was carried-out for the UK as a proof-of-concept. The grid components selected for evaluation were: gas generation plants, nuclear reactors, solar photovoltaic panels, transformers, overhead lines and peak daily demand; the weather variable studied was ambient temperature. Potential impacts were calculated using maximum daily and average ambient-temperature datasets. These were recovered from UKCP18 models and associated with the linear impact correlations extracted from the review. Resulting maximum daily impacts are delivered at a 60 km^2 resolution for the 2030-2050 and 2080-2099 periods; they were derived based on the p90 of ambient temperature values from 28 model members.

This impact calculation and subsequent visualisations showed that impacts are projected to be more severe in the higher-temperature regions of the country's southeast. In those areas, maximum daily temperatures could reach 45°C by the end of the century, causing the generators' capacity to fall by up to 22% of rated for gas plants, 25% for nuclear plants and 9% for solar panels. Meanwhile, transformer load capacity could be reduced up to 25% of rated and overhead lines lose up to 28% carrying capacity. On maximum temperature days, peak demand could be increased up to 70% of its average spring value.

The generation capacity losses could theoretically be absorbed by the UK's capacity margin. However, factoring in the combined impacts of lowered transmission capacities and increased peak demand could potentially bring the National Grid to failure. Important adaptation efforts are expected to be required in order to avoid blackouts and component loss-of-life by mid- and end-of-century.

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Acronyms

AC Air Conditioning. 36

AR6 Sixth Assessment Report. 12

CCGT Combined Cycle Gas Turbines. 33

CCRA Climate Change Risk Assessment. 14

CMIP5-13 Coupled Model Inter-Comparison Programme. 22

DOs Data Objects. 37

ESs Electricity Systems. 11

GCOs Grid Component Objects. 37

GCRT Power Grid Climate Resilience Tool. 41

GCs Grid Components. 11

GE General Electrics. 33

GHG Greenhouse Gas. 12

IEASDS International Energy Agency Sustainable Development Scenario. 11

IEEE International Electrical and Electronics Engineers. 19

IPCC Intergovernmental Panel on Climate Change. 12

LOLE Loss Of Load Expectation. 18

LOLF Loss Of Load Frequency. 18

MED Major Event Days. 19

mSI monocrystalline silicon. 34

NAP National Adaptation Programme. 14

NetCDF Network Common Data Form. 47

NGET National Grid Electricity Transmission. 32

PPE-15 Perturbed Parameter Ensemble. 22

pSi polycrystalline silicon. 34

PV Photovoltaics. 17

RCP Representative Concentration Pathways. 22

SAIDI System Average Interruption Duration Index. 18

SAIFI System Average Interruption Frequency Index. 18

SDG Sustainable Development Goal. 11

SRES Special Report Emission Scenario. 22

UKCP UK Climate Projection. 14

USA United States of America. 12

USD US dollars. 12

WV Weather Variables. 31

Chapter 1

Introduction

This thesis work provides the frameworks and tools necessary to study the impacts of climate change on Electricity Systems (ESs). It describes literature research work and the analysis conducted of potential impacts on the UK electricity grid as well as the web application developed to visualise them.

After arguing why impacts of climate change on electricity systems should be studied, this section outlines the work's scientific direction, aims and scope. The following Chapter 2, 'Background', provides the frameworks necessary to describe the analysis carried out. It defines the terms and methodologies provided by resilience studies and those inherited from climate sciences. It also describes the studied system and reviews works addressing weather impacts on Grid Components (GCs). The 'Data Collection and Analysis Methodology', Chapter 3, details the climate data and impact-function choices as well as the approach used to calculate high-temperature impacts on individual GCs across the UK. The 'Results, Discussion and Conclusion', Chapter 4, summarises the functionalities of the developed program and visualisation tool. It subsequently displays and comments on the obtained impact statistics per studied GC and associated impact maps. Finally, it discusses the results in the broader context and concludes providing an outlook on their possible implications.

1.1 Electricity grids as critical systems

Electricity systems are essential to human prosperity. In 2018, they carried almost 20% of the world's final energy consumption, potentially reaching 28% in 2040 under the International Energy Agency Sustainable Development Scenario (IEASDS) [1]. Today, close to 90% of the global population has access to electricity - albeit with significant disparities between Sub-Saharan Africa and the rest of the world [2]. While the United Nation's Sustainable Development Goal (SDG) number 7 sets the objective to "ensure access to affordable, reliable, sustainable and modern energy for all" [3], countries aiming for Net Zero by 2050, like France and the UK, are starting to transform their networks profoundly. Current power systems will evolve dramatically as electrification increases and power generation shifts towards low-carbon sources. France, for example, plans on doubling renewable power generation in the next 15 years. Meanwhile, the UK set off to fully decarbonise electricity by 2035 with a parallel four-fold increase in production [4, 5].

Switching from fossil fuels to intermittent renewables carries challenges in providing reliable and affordable electricity, even more so with growing demand. As per definition in [6], power systems are a collection of "physical or intangible assets, whose destruction or disruption would seriously undermine public safety, social order and the fulfilment of key government responsibilities", fitting them under the "critical systems" category. These critical systems are on the track of ambitious and challenging transformation, and yet, composed chiefly of over-the-ground assets, they are highly vulnerable to weather events.

1.2 Weather factors and electricity-grid failure

A recent and emblematic example of the wide-scale impacts of extreme-weather events on electricity grids is the Texas 2021 Power Crisis [7]. As a series of severe winter storms fell on the area and its neighbouring states in February 2021, 69% of the state population lost power for an average of 42 hours, with half of the population losing access to water supply as well, due to inability to power pumps. The death toll reached several hundreds of lives, while financial impacts reached hundreds of billions of USD [7].

Even more recently, the 2022 summer heat waves created severe temperature increases across Europe, the Americas, China, India, and Australia. The UK, in turn, paid record prices for energy imports on July 20th – £9,724.54 per MWh, more than 5,000% the typical price - to avoid partial blackouts in London. Electricity imports from Belgium were secured to address a combination of a limited line carrying capacity across the UK and increased demand in the South of the country [8]. In parallel, the UK issued summer capacity warnings – rare in the past – all the more frequently [9, 10].

Increases in weather-related grid disturbances are not anecdotal. The United States of America (USA) reports a sharp increase in weather-related outages between 1992 and 2002. Over that period, they represented over half of all outage events and 90% of those affecting over 50,000 people [11]. Electricity systems are vulnerable to weather events, and failure to build adaptation of distribution and transmission assets can significantly raise financial risks. A 2020 study evaluated potential economic impacts under two Greenhouse Gas (GHG) emission scenarios and three adaptation strategies in the US. It found that no adaptation could cost up to 13 billion US dollars (USD) per year by the end of the century under the moderate-emissions scenario and double that under the high-emissions scenario. Adaptation efforts have a crucial role to play in protecting power systems and reducing running costs, as they could halve those expenses, according to the study [12].

1.3 Climate change and extreme-weather event occurrence

1.3.1 Future global climate - an outlook

Regardless of the modelled emission scenario – from an immediate stop of anthropogenic GHG emissions to the 'worst-case' scenario – extreme weather will increase both in frequency and severity. The 'Impacts, Adaptation and Vulnerability' section of the Intergovernmental Panel on Climate Change (IPCC) Sixth Assessment Report (AR6) shows that climate change has adverse impacts on infrastructure across all regions of the world already today [13]. As global temperatures continue to rise and the water cycle becomes heavier, preventing power outages will increasingly demand careful assessment of the risks posed by weather factors and the implementation of appropriate resilience strategies.

On the global scale, the IPCC's AR6 report shows that most of the world's regions are experiencing an increase in hot-temperature extremes, compared to the 1850-1900 period, with a global average temperature increase of $1\,^{\circ}$C [13]. Surface temperatures will continue to rise until mid-century, regardless of the explored mitigation scenario. Long-term modelling (2081-2100) shows dramatic increases in average temperature across emission scenarios: $1\,^{\circ}$C - $1.8\,^{\circ}$C under the SSP1-1.9 scenario and $3.3\,^{\circ}$C - $5.7\,^{\circ}$C under SSP5-8.5, compared to the 1850-1900 period. The likelihood of exceeding either $1.5\,^{\circ}$C or $2\,^{\circ}$C of warming by the end of the century is reported as extremely high unless GHG emissions are seriously lowered in the next decades. Such warming directly correlates with increases in frequency and intensity of extreme-weather events, like land and marine heat waves, heavy rainfall, droughts, and storms.

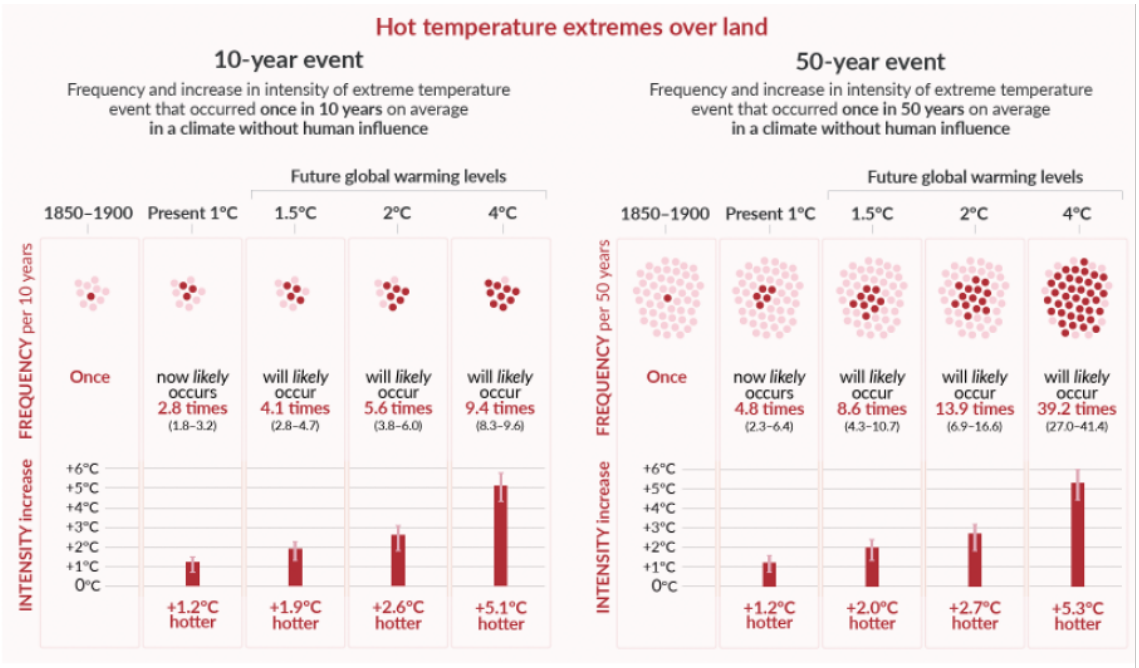


Figure 1.1: Frequency and increase in intensity of extreme-temperature events that occurred once-in-10-years (left) and once-in-50-years (right) on average in a climate without human influence - IPPC AR6, 2021.

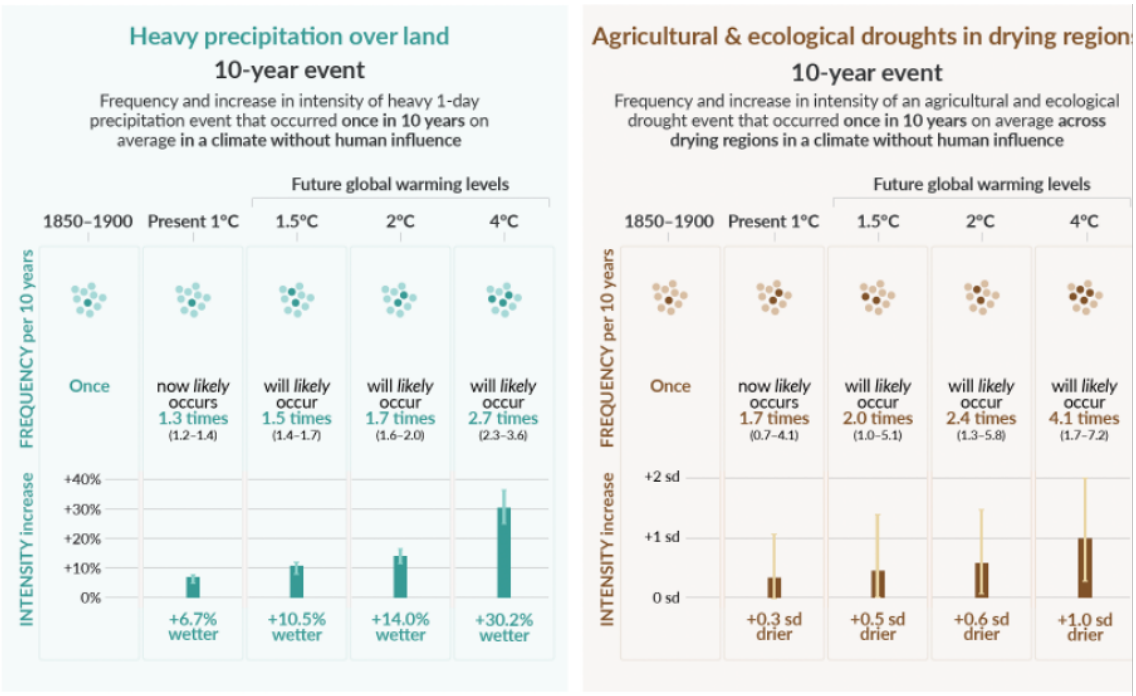


Figure 1.2: Frequency and increase in intensity of heavy 1-day precipitation events (left) and of a drought event (right) that occurred once-in-10-years on average in a climate without human influence - IPPC AR6, 2021.

The increase of such extreme-weather events is schematically illustrated in Figure 1.1 and Figure 1.2. Figure 1.1 shows the frequency and increase in intensity of extreme-temperature events that occurred once-in-10-years and once-in-50-years on average in the period 1850-1900, that is, in a climate without human induced radiative effects. It also shows expected future warming levels and their impact on temperature increase. Similarly, Figure 1.2 shows the frequency and increase in intensity of heavy 1-day precipitation events as well as of agricultural and ecological drought events, that occurred once-in-10-years, on average, in the period 1850-1900, together with the expected future warming levels and their projected impact on wetter or drier weather respectively.Values of maximum wind speeds and tropical cyclone intensity grow in average value globally. Daily rainfall, for example, is described as potentially growing by 7% for every degree of warming on a global average. These reports clearly stress the growing risks threatening infrastructures and systems, like electricity networks across the globe. The required resolution will nevertheless be much higher for practical risk assessments. Network scale resolutions will be needed to study equilibriums between generation and demand, while assessments of extreme-weather impact call for regional to local scales. Obtaining projection data on such scales is possible but carries additional uncertainties.

1.3.2 Future climate in the UK - an outlook

The latest issue of the UK Government's Climate Change Risk Assessment (CCRA) report, based on data from the 2018 UK Climate Projection (UKCP), outlines the increase in severe weather across the UK by 2050 and by the end of the century. Results show that hot summer days will be more frequent. Events such as the 2018 heat wave, reaching 35 °C, are already 12-25% more likely today, and will be 50% more likely by 2050 [14]. Precipitation is likely to decrease in summer by a maximum of 47%, and to decrease in winter up to 35%. Rainfall events show a simultaneous increase in event severity for both seasons. Precipitations associated with one-in-every-two-year extreme rainfall events are set to increase by a third, while lying snow levels are projected to decrease by almost 100% all over the UK, implying severe disturbances of water cycles and year-long availability [14]. Meanwhile, sea level rise in the high-emission scenario is estimated to reach 0.5-1.2 m in the London area and 0.3-0.9 m in the Edinburgh area by the end of the century, with implications for flooding of local infrastructure.

These results outline that extreme-weather events will increase in frequency and severity even with serious climate-mitigation efforts. Adaptation plans are therefore crucial and must include the 'worst case' climate-change scenarios to build the resilience of critical systems sufficiently against rare but severe events.

1.4 Resilience and adaptation in UK national strategies

In the context of climate-resilience studies, it is essential to distinguish between climate mitigation and adaptation. Climate mitigation is a term denoting the set of efforts aiming to reduce the greenhouse gas levels in the atmosphere, either by lowering emissions of those gases or enhancing 'sinks' allowing for their natural or anthropic capture and storage. Reducing their levels lowers their radiative effect on the atmosphere, which causes increases in global temperatures and transforms the Earth's climate [15]. Adaptation efforts, in contrast, involve adjusting to occurring and projected climate change, reducing the risks from its harmful effects [16]. Albeit both related to climate change, mitigation and adaptation strategies can interact positively or have trade-offs [17]. In the context of ESs, climate-resilient designs could have mixed effects on mitigation. The extensive deployment of energy- and resource-intensive battery packs, could, for example, stabilise the grid but at a high emissions price. However, it could also lead to overall positive mitigation, as protected networks need less maintenance and replacing.

The power and network industries have extensively studied asset adaptation to weather in the specific context of electricity systems. As reported in Section 2.4.3, there is a vast collection of scientific and industry publications exploring methodologies for resilience assessment, system or component fragility and advising on adaptation strategies. However, a dedicated UK government planning document, laying out a roadmap for resilience and adaptation efforts, is still to be published. The UK's 2018 National Adaptation Programme (NAP), written as a response to the CCRA

and meant to highlight key actions for the next five years, dedicates a limited part to the energy sector, which can be a starting point. It mentions flooding risks and the importance of power network companies' reporting; however, it needs to be complemented by highlighting adaptation measures and designing operational roadmaps [18, 19].

1.5 Problem definition, aims and scope

1.5.1 Problem definition

Electricity systems are critical and vulnerable, with substantial risk to populations in case of widespread or long-term failure. Industry risk assessments underline the need for long-term planning designed by industry stakeholders, grid operators and regulators to ensure the resilience of existing but also newly-installed assets [20]. Overall, there is a pressing need for governmental planning to guide adaptation efforts for the national electricity or energy sectors. The difference between how this subject is approached in the academic and industry publications, on one side, and the national planning on the other, could be the result of a lack of intuitive content allowing fully apprehending the magnitude of climate risks on electricity systems or assisting stakeholders in the prioritisation of resilience efforts.

As commented by Dr Hebner, Director of the Center for Electromechanics at UT Austin, after the Texas Power Crisis - *"The real outstanding question nationally is, given climate change, how much more extreme weather should we be protecting against"* [7]. This thesis work can contribute to such outstanding issues by providing an easy-to-use interactive tool allowing intuitive impacts visualisation for different weather conditions.

1.5.2 Aims and objectives

This work aims to develop and deliver an intuitive, easy-to-use, tool to visualise climate impacts on electricity grids, based on researched literature and educated extrapolations of the future. The first objective is to provide a functional proof-of-concept, using estimates of such climate impacts in the UK. This first released version focuses on high-temperature impacts on grid components around mid- (2030-2050) and end-of-century (2080-2099). Its main objectives are the following: 1) Gather and categorise works describing the impacts of extreme weather on grid components; 2) Provide an interactive tool allowing the visualisation of weather impacts on electricity systems; 3) Evaluate the scale of high temperature impacts in the UK. Secondary objectives include building a base for geographically broader analysis and visualisations, and aiding prioritisation between resilience efforts.

1.5.3 Scope

The following analysis focuses on the impact of temperature elevation on electricity grid components. For simplicity of modelling and clarity of results, it is component-specific and not system-wide. Electricity grid support systems, such as transportation, telecommunications, upstream fuel and materials production and supply are considered intact despite weather pressures. Impact is evaluated with fragility functions that are specific to each grid-component and weather-variable pair. Results do not account for interactions between components or simultaneously occurring weather pressures. Impact functions are applied to location-specific climate projection data. These capture effects on hypothetical present-day standard assets without accounting for the actual location of those assets in the UK. Future technological development, operational aspects, or population growth are not accounted for in the impact assessment.

Chapter 2

Background

This section provides the frameworks underlying the analysis in Section 3. Firstly, Section 2.1 defines the scope, terms and methodologies usual in grid resilience studies. Section 2.2 describes the elements of the UK's ES under study in this work, while Section 2.3 reviews available climate projection data. Finally, Section 2.4 provides methodologies for impact quantification and summarises the literature reporting on GC weather impacts, justifying subsequent scope choices.

2.1 Key frameworks and definitions

2.1.1 Electricity systems

The collection of elements referred to as the "electricity system" in this work is the one illustrated in Fig. 2.1 per [21]. Its boundaries start with generation facilities: thermoelectric (gas and nuclear) powerplants, hydroelectric plants, wind generation and solar Photovoltaics (PV) farms. It follows the electricity transportation route through the transmission and distribution systems: step-up and step-down transformers, transmission poles, aerial and underground lines, distribution lines, and substations. The boundaries include final consumption sites and customers in the system, as their behaviour is closely linked to final electricity demand and, thus, relevant to determining the load on the system. Some of the reviewed resilience studies included upstream processes like oil and gas exploration and supply chains [22]. Including them allows better energy security analysis. It also addresses the interdependencies between sectors, like gas and electricity, which seem to increase with the growing need for flexibility in decarbonising grids [23]. Adding such interdependency layers, nevertheless, increases the system complexity exponentially, limiting the number of works with such a broad scope. Most resilience-quantification methodologies, as described later in Section 2.4.1, bring the electricity system, only, into the scope; excluding from their assessments fuel supply chains, production and installation of components, or wider socio-economic factors; a boundary also applied in this work's methodology.

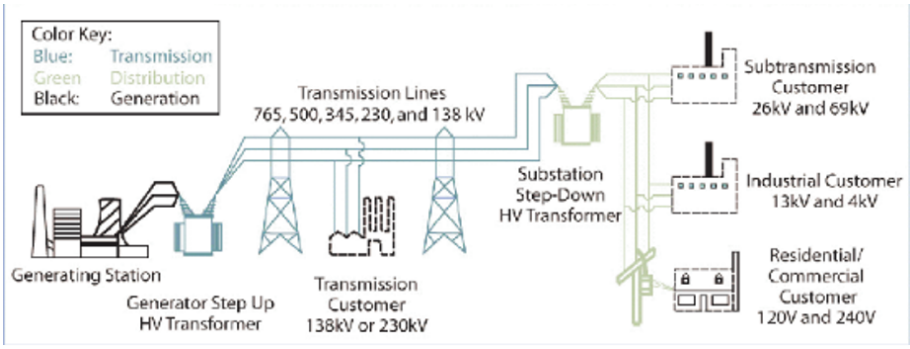


Figure 2.1: Electricity System components and power flows - Campbell, 2012.

2.1.2 Resilience and reliability

Electricity grids are designed under the imperative of preserving operations at maximum transmission and supply capacity even while having one component in a failed or maintenance state. This principle is described as "N-1" security. However, it can grow to include one component outage and one in maintenance at any given time ("N-2" security) for more sensitive networks [24]. Events, like adverse weather, can cause the failure of multiple components simultaneously, resulting in an "N-k" outage, potentially beyond built-in redundancy [25]. Such outages can also occur for non-weather-related reasons, falling under the scope of resilience studies. Identifying disruptions under 'normal' operating conditions and separating them from significant 'extreme-weather' disruptions is, thus, essential when assessing historical outage data.

In its broad definition, a system's 'resilience' describes its capacity to anticipate disruptions, withstand them, recover, and adapt to face future hazards better [26]. Studying ESs climate resilience examines those behaviours under extreme-weather conditions, usually as functions of weather factor intensity, grid exposure and grid vulnerability [26]. In this context, resilience can improve by mitigating exposure and vulnerability or increasing network flexibility and redundancy [27]. These concepts relate to the already widely-established notion of grid reliability. Defined by its focus on simple and short interruptions with straightforward recovery actions, reliability is used to study networks under normal operating conditions, explicitly excluding extreme-weather impacts by, for example, selecting faults shorter than 3 or 5 minutes [26]. The two definitions are interconnected but can cover different aspects of network behaviour.

Metrics of electricity-network performance traditionally measure interruption frequency, duration, and the number of impacted customers. Indices often used in reviewed works are (a) the System Average Interruption Duration Index (SAIDI) accounting for the average yearly period during which power supply to customers is interrupted, (b) the System Average Interruption Frequency Index (SAIFI), measuring the number of interruptions suffered by customers per year, (c) the Loss Of Load Frequency (LOLF), measuring the number of interruptions in a given time frame, and (d) the Loss Of Load Expectation (LOLE), which measures the number of hours or days per year where production will not meet demand on the network [25, 27, 28]. These indicators can also be used to identify significant interruptions due to extreme weather but fail to incorporate any notions of anticipation, recovery or adaptation of the network [25]. Such metrics, nevertheless, provide a foundation to incorporate the fundamental adverse impacts of extreme weather on electricity systems, outages, their scale and duration into resilience studies.

2.1.3 Characterising extreme-weather events

Extreme-weather events are characterised by their low-occurrence probability and high potential for inducing extensive damage. They can be identified based on probability of occurrence, as for once-in-50-year events. Measurements of an observational variable can be compared to a set threshold to identify an extreme event; for example, a wind-speed measurement above a threshold value set independently, or a specified level of caused damage. Nevertheless, characterisations of the same event can vary significantly in geographical and time scales between different works [29].

To solve identification issues like these in climate studies, Cattiaux and Ribes argue for the definition of a single extreme-weather event based on the minimisation of its occurrence probability [29]. Their approach spawned from the observation that estimations of occurrence probabilities for weather events vary depending on chosen geographical or time scale parameters. In their case these were: the choice of calendar, annual or local maxima temperatures when characterising the 2003 European heat wave event. They, thus, recommend consistently defining single weather events with the spatiotemporal scale that minimises their occurrence probability. This definition provides a clear, reproducible, and universal description for each event and allows for automatic screenings. It may, nevertheless, require considerable computational powers. It may also force risk assessments to work within the very small probability domain, at the risk of underestimating actual occurrence probabilities and necessitating appropriate mathematical tools.

In the context of power-network operations, the International Electrical and Electronics Engineers (IEEE) guidance defines the "Major Event" category as a set of conditions causing electricity systems to "exceed reasonable design or operational limits" [30]. Major Event Days (MED), in turn, are defined as days for which SAIDI values exceed a previously set threshold. According to the guidance, those data points typically include extreme-weather events and should be treated outside reliability studies. Event identification can, thus, be based on grid impacts rather than the weather observations constituting them. Such an approach can be useful when focusing on power systems since it allows the identification of significant events by analysing historical grid operations data; without relying on climate science's definition of extreme-weather events. Associated historical weather data can later be used to verify results. The limitations of such an approach are nevertheless clear, as it can only apply to historical events or weather projections associated with related SAIDI projections. The results will also be very network specific, as SAIDI values depend on network structure and its current capacity to withstand extreme weather. It also means that as networks adapt to face future events better, potentially reducing SAIDI values, the threshold characterising extreme-weather events will change. Resilience studies identifying extreme weather based on impact, thus, need to justify the coherence of this definition across time and systems.

Grid-resilience assessment works reviewed under [26] mostly seem to use approaches related to the IEEE's guidelines, defining weather events from grid impact rather than the other way around. Some introduce threshold values for weather variables based on previous works showing that those thresholds are significant in terms of associated damage (e.g. wind speeds exceeding 17 m/s in [31]). These can be accompanied with standard time ranges defining the event as a window around the occurrence of the threshold value. The variety of definitions found in publications suggests that it is common practice to characterise extreme-weather events in specific ways, relevant to the particular study, rather than relying on a universal definition. In the case of ES impact assessments, a simplified methodology - used in this work - for extreme-weather event characterisation could rely on pre-set threshold values for weather variables, chosen according to the vulnerability of the grid components under study. Finally, an alternative simplified characterisation, relying on impact, could be used defining significant events as those with power outages longer than a given time limit (above 5 minutes for [26]) or those impacting at least a certain number of customers (over 50,000 for [21]).

2.2 The UK's electricity network

The UK's National Grid operates at 50 Hz and covers England and Wales with connections to Northern Ireland, Ireland and mainland Europe. Generation is ensured, in part, by close to 460 thermal power stations. Electricity transmission is ensured by 7,200 km of overhead lines, 2,200 km of underground lines, 346 substations and around 90,000 pylons, connecting 30 million homes and businesses across the UK and Ireland [32, 20]. Main overhead transmission lines, a mix of 275 kV and 400 kV, can be seen in Fig. 2.2 with the National Grid lines in yellow [33].

In 2019, power generation was mainly covered by Gas plants (41%), Wind and Solar installations (24%), Nuclear power (17%) as well as Biofuel combustion (11%). The complete generation landscape is drawn in Fig. 2.3 based on National Statistics data [19].

Plant locations and installed capacities are illustrated in Fig. 2.4. Coal, Gas and Nuclear plants are located across Great Britain, although most are close to bodies of water to secure cooling. Nuclear plants, in particular, are all found on the coastline. The 1 MW capacity minimum of this census excludes rooftop Solar PV. Despite its absence, it is clear that solar and wind installations are more distributed than thermal generation plants. Solar PV is primarily found in the south to capture greater insulation levels. Rainfall patterns and landscapes favoured hydroelectric generation in Scotland and Wales, while wind farms are usually located along coastlines and on higher ground to capture stable winds. Each location choice has implications regarding climate risk exposure. Coastal areas are more exposed to sea-level rise, while the country's south will suffer more from temperature elevations.

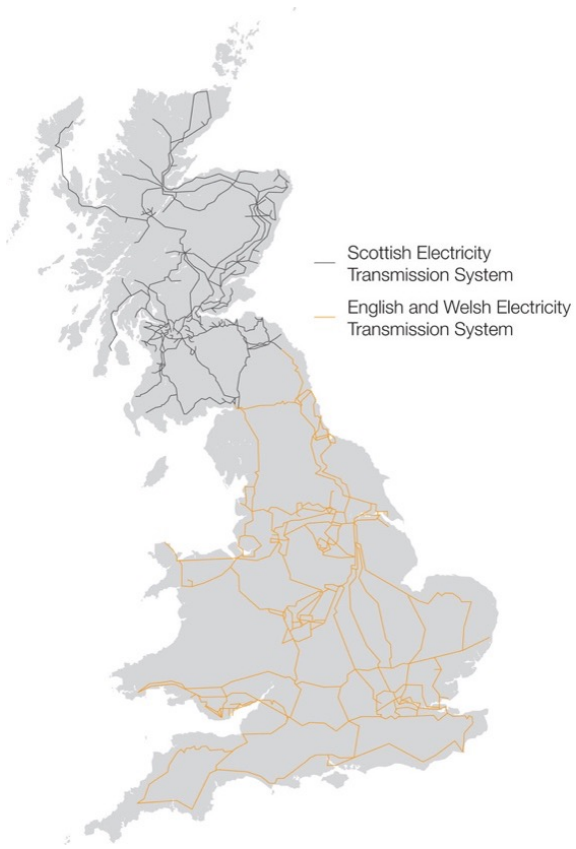


Figure 2.2: Map of the UK power transmission line network - © National Grid UK, 2021.

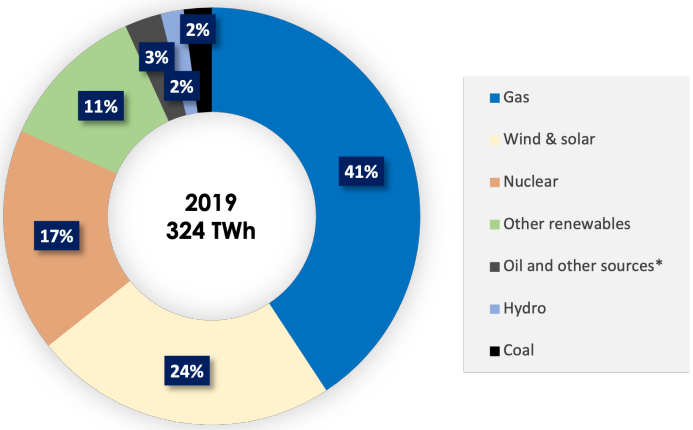


Figure 2.3: Electricity generation by source in the UK in 2019 (TWh) - drawn from National Statistics data.

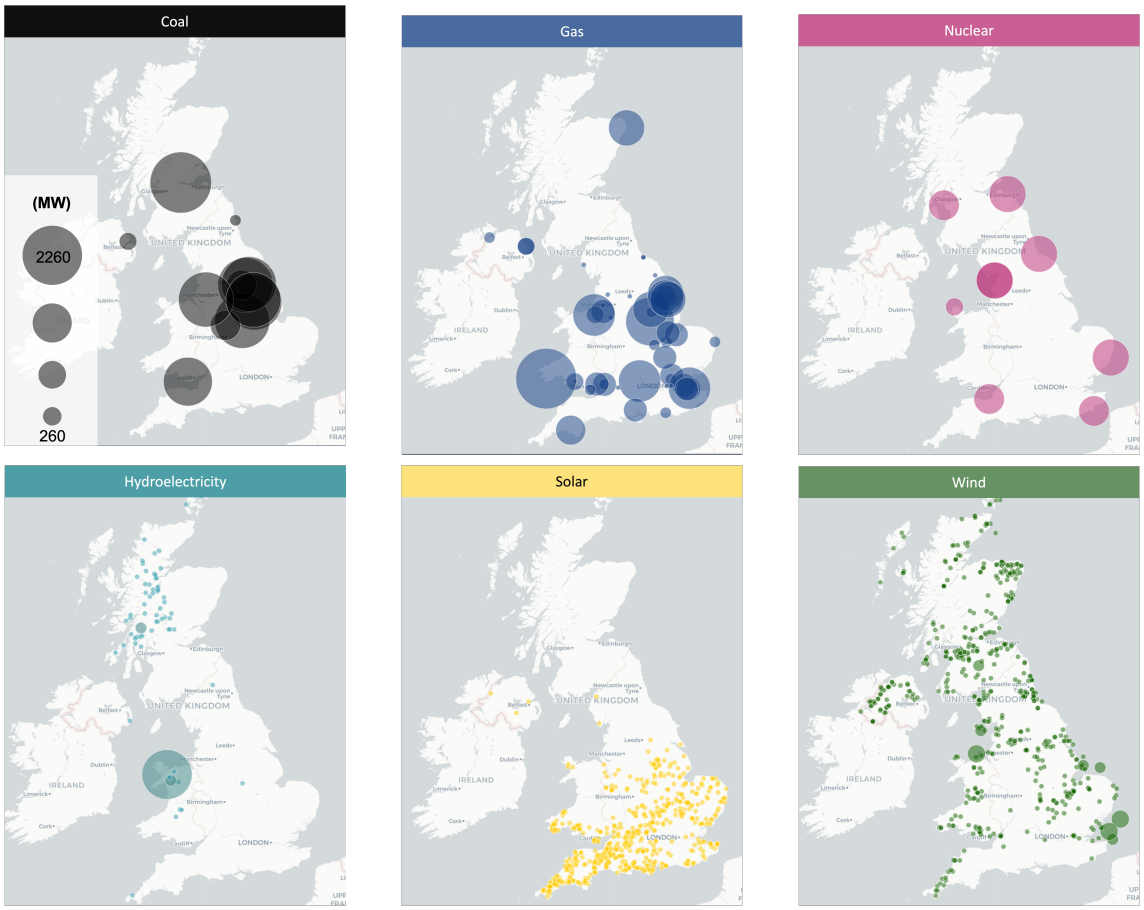


Figure 2.4: Installed power-generation capacity in the UK for different sources in MW. Rooftop solar PV is excluded from this census - Rosamund Pearce, Carbon Brief and CartoDB, 2022.

2.3 UK climate projections

2.3.1 Definitions

All definitions were written using the UKCP User Training material as reference [34].

Climate models allow scientists to replicate the mechanics of climate by numerically solving equations through time. These equations describe the energy balance, motion and thermodynamics of earth system components, including the atmosphere, oceans, land or vegetation systems. Solutions are computed over regular time intervals (typically 10-30 minutes) over grid cells whose size sets the simulation resolution. Climate models typically rely on emission scenarios to set initial conditions and key variables.

Emissions scenarios include hypotheses describing future emissions or atmospheric concentrations of greenhouse gases and aerosols. Scenarios can rely on assumptions concerning future socioeconomic or technological development. An important collection of scenarios is called the Representative Concentration Pathways (RCP), used in the IPCC's fifth report [15]. Underpinning these scenarios are a set of assumptions on radiative forcing; rather than socioeconomic storylines as with the previous Special Report Emission Scenario (SRES). Four pathways were developed – RCP2.6, RCP4.5, RCP6.0 and RCP8.5 – with each index indicating a level of forcing in Watts per metre square (W/m^2) compared with a pre-industrial reference. RCP8.5 represents a very high baseline emissions scenario. It should not be considered an extension of current trends or a likely "business as usual" scenario. It should rather be regarded as a high-emissions "no mitigation policy" scenario with the same likelihood as other no-policy baseline scenarios.

Climate projection refers to the resulting simulation response of a climate model given a chosen emission scenario.

Model members represent the different versions of the future or "realisations" that climate models can produce. The UKCP Global (60 km) model contains 28 members. Of those, 15 members of the Perturbed Parameter Ensemble (PPE-15), originated from the same model, the Met Office Hadley Centre global climate model (HadGEM3-GC3.05), through parameter perturbation. The remaining 13 were generated from different models established by centres internationally, as part of the Coupled Model Inter-Comparison Programme (CMIP5-13). As each realisation of the future is equally probable as another, sets of members provide the broadest range of outcomes and allow uncertainty assessing.

2.3.2 UK climate projection frameworks

The UKCP18 set of models includes Global (60 km grid cells), Regional (12 km grid cells) and Local (2.2 km grid cells) models, each spanning different geographical areas and providing its particular benefits. A more detailed description of each model is provided in Table 2.1 extracted from the Met Office Hadley Centre guidance [35].

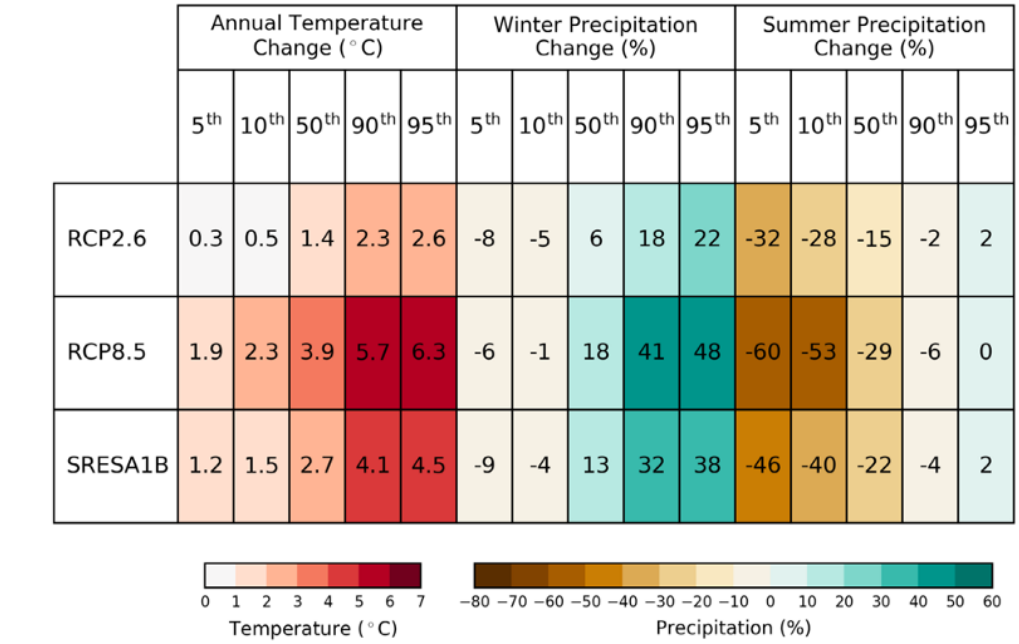
Table 2.1: The UKCP18 framework, models and their characteristics - Meteorological Office Hadley Centre, 2018.

	Global Climate Models	Regional Climate Model	Convection-Permitting Model
Spatial resolution	60km	12km	2.2km
Geographical coverage	Global	Europe	UK
Scenario	RCP8.5, RCP2.6	RCP8.5	RCP8.5
Key benefits	Long time series (1899-2100) Multiple climate models	Enhanced spatial detail Improved extremes component	Enhanced spatial detail Improved extremes component Improved representation of intense storm events
UKCP product	UKCP Global (60km) Derived Projections	UKCP Regional (12km)	UKCP Local (2.2km)

The subsequent climate projections originate from the UKCP Global (60 km) model and its associated probabilistic projections. As summarised in Table 2.2, different scenarios entail varying temperature and precipitation changes across time, with modelling extending from 1899 to 2100. The summarized values provide an outlook of average annual change for the 2080-2099 period. Each percentile indicates the percentage of data found below its associated value. Annual average values are thus found under the 50th percentile, with other percentiles representing more "extreme" and more "rare" values of that variable. Extreme-weather events reside at the tail ranges of percentiles, where uncertainty is the greatest. This tendency illustrates one of the main reasons they are hard to study: the more extreme projected events are, the lower their probability is and the greater the associated uncertainty. In order to capture extreme events with a reasonable degree of likelihood, this work focuses on the light-shaded temperature areas found between the 10th and 90th percentiles in Fig. 2.5.

Table 2.2: Projected change in UK temperature for the 2080-2099 period relative to 1981-2000, based on probabilistic projections of emission scenarios RCP2.6 (left) and RCP8.5 (right)- UKCP18, 2018.

	Annual Temperature Change (°C)					Winter Precipitation Change (%)					Summer Precipitation Change (%)				
	5th	10th	50th	90th	95th	5th	10th	50th	90th	95th	5th	10th	50th	90th	95th
RCP2.6	0.3	0.5	1.4	2.3	2.6	-8	-5	6	18	22	-32	-28	-15	-2	2
RCP8.5	1.9	2.3	3.9	5.7	6.3	-6	-1	18	41	48	-60	-53	-29	-6	0
SRESA1B	1.2	1.5	2.7	4.1	4.5	-9	-4	13	32	38	-46	-40	-22	-4	2



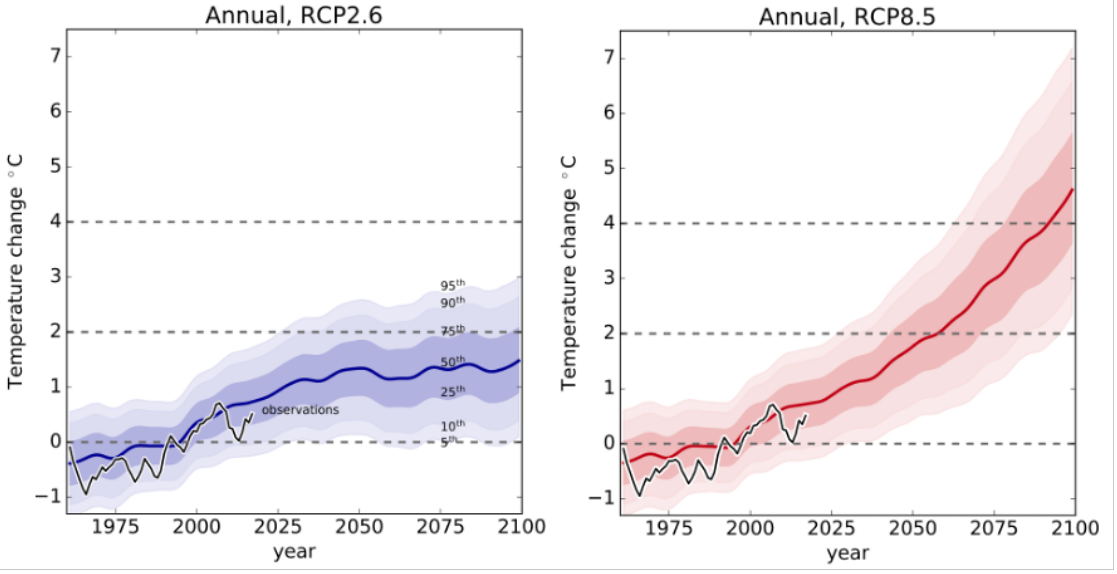


Figure 2.5: Projected change in UK temperature and precipitations for the 2080-2099 period relative to 1981-2000, based on probabilistic projections - UKCP18, 2018.

2.4 Studying climate impacts on electricity grids

2.4.1 Methodologies for fragility-curve derivation

The vulnerability of network elements can be quantitatively described using functions relating weather-variable values to impact levels. These can take the form of fragility curves or damage functions, relating a probability of failure to hazard intensity. Fragility curves can express the probability of failure of an overhead line as a function of wind speed as does the generic curve reproduced in Fig 2.6 according to [36]. They can also relate a performance metric to the weather variable, such as the maximum capacity of a transformer in terms of air temperature [27].

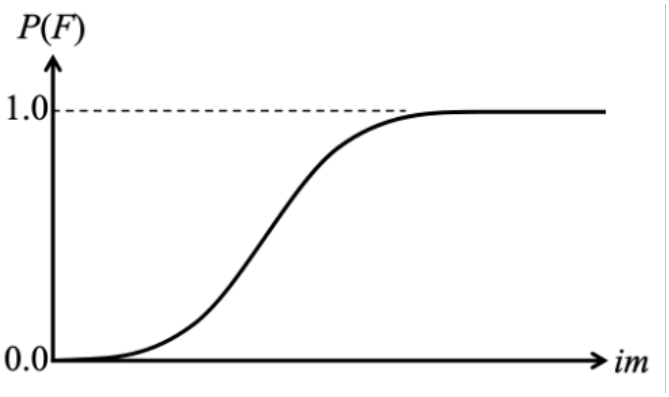


Figure 2.6: Generic fragility curve of the failure probability of a component, $P(F)$, as function of the observed variable intensity, im - Haukaas, 2020.

Four different methods exist to derive impact functions for individual components or systems: judgemental, empirical, analytical and hybrid [27, 31]. The judgemental approach relies on engineering and expert opinions and, thus, often carries experience biases. It nevertheless has the advantage of needing little extra data collection or processing and, therefore, remains a possibility when data is lacking. The empirical method relies on a set of observations matching the studied variable with impact levels on the studied system. This usually uses historical data as it is expensive and time consuming to test components to physical failure in the lab. Nevertheless, such an approach depends on observational data availability. Additionally, the resulting fragility curves are specific to the studied system, meaning they have low transferability to other systems. Finally, the analytical method relies on developing structural models, describing the considered system, and its limit states. This last approach can be the most precise but is limited by potential modelling shortcomings and computational time. Note that it is possible to consider an entire network as a unique grid element and develop a fragility curve based on extensive fault datasets without attempting to provide an analytical description of the system as a whole [31].

2.4.2 Methodologies for system-resilience studies

Multiple methodologies aim to quantify resilience at the system level for electricity networks. Metrics, like network impact, can be mathematically expressed using previously defined grid states [26, 37]. As described in the study, total impact can be defined using grid functionality states as follows:

$$I = \int_{t_E}^{t_R} [Q_N - Q(t)]dt \tag{2.1}$$

where I is the total impact,
t_E and t_R are the event start and restoration times respectively,
$Q(t)$ is the grid functionality at time t,
Q_N is the full grid functionality.

Grid functionality $Q(t)$ is itself defined as follows:

$$Q(t) = \sum_{n}^{N} (\rho_n \times q_n) \tag{2.2}$$

where ρ_n is the node status at time t, either operational (= 1) or failed (= 0),
q_n is the load capacity of node n.

Determining the status of each node (component or line) can be done following system-specific methodologies. The simplest examples are either comparing a node's physical parameters to its physical limit values, or using automated power flow analysis and excluding elements operating above their rated values for current or voltage [26]. This last method allows incorporating cascading electrical failures into the model. More complex methodologies, relying on statistical approaches, calculate the element's failure probability and compare it with a random number selected between 0 and 1. If the random number is higher than the computed probability, the element is considered functional; otherwise, it is considered tripped [38]. This last step incorporates event randomness into the model, but little justification is given regarding its physical basis or how realistic the model is.

Each one or a combination of these methods could be used to evaluate failure probabilities for individual components and aggregate them into grid impact metrics, allowing comparisons across weather conditions and networks. These methods potentially pave the way for simplified assessments across geographical areas and weather conditions (historical or projected). Such an approach shows variation in impact and, therefore, priorities for each configuration.

2.4.3 Review of climate-impact studies on electricity systems

Reference papers reporting on the impact of weather pressures on specific grid components were categorised according to studied components and weather variables. Table 2.3 shows the resulting categorisation and Table 2.4 provides the key for cited sources. An added sign (Q) indicates those providing a quantified relationship between observed variable levels and impact on studied components. The potential implications of climate change and weather hazards on grid components are summarized in Table 2.5. The main identified risk areas involve thermoelectric generation and transmission assets which seem to be particularly vulnerable to temperature effects, water availability and severe weather hazards like flooding or storms. Section 3.1.1 describes the source selection and categorisation process in fuller detail.

Table 2.3: A review of studies relating weather pressures to specific grid component damage.

	Component \ Weather Variable	Heat Waves	Ambient Temperature	Water Temperature	Temperature Amplitudes	Wildfires	Draught	Wind speed	Electric storms	Water availabilty	Precipitation	Sea level Rise	Flood	Cold Waves / Ice storms	Snowfall	Vegetation
Generation Technologies	Thermoelectric Generation - Gas	- Abi-Samra, 2011 - Añel, 2017 (Q) - Dumas, 2019 (Q) - Lubega, 2018	- Aivalioti, 2015 - Dumas, 2019 (Q) - GE 2001 (Q) - Sathaye 2013 (Q) - Zamuda 2013 (D)	- Aivalioti, 2015 - Dumas, 2019 - Zamuda, 2013			- Dumas, 2019 - Lubega, 2018	- Arab, 2015		- Dumas, 2019 - Lubega, 2018 - Zamuda, 2013	- Sathaye, 2013	- Dumas, 2019 - Sathaye, 2013	- Dumas, 2019 (Q)	- Abi-Samra, 2011		
	Thermoelectric Generation - Nuclear	- Abi-Samra, 2011 - Añel, 2017 (Q) - Dumas, 2019 (Q) - Lubega, 2018	- Aivalioti, 2015 - Dumas, 2019 - Linnerud, 2011 (Q) - Sathaye, 2013 - Zamuda, 2013	- Abedi, 2019 - Aivalioti, 2015 (Q) - Attia, 2015 - Añel, 2017 (Q) - Dumas, 2019 (Q) - Zamuda, 2013			- Lubega, 2018	- Arab, 2015		- Añel, 2017) - Lubega, 2018 - Zamuda, 2013	- Sathaye, 2013	- National Adaptation Program, 2018 - Sathaye, 2013 - Zamuda, 2013	- Dumas, 2019 (Q) - National Adaptation Program, 2018 - Zamuda, 2013			
	Hydroelectric Generation	- Aivalioti, 2015 - Dumas, 2019 (Q) - Qin, 2020	- Dumas, 2019 - Zamuda, 2013 - Kopytkovskiy, 2015 - Madani, 2013 - Mehta, 2011				- Qin, 2020			- Dumas, 2019 - Gjorgiev, 2017 - Lubega, 2015 - Vliet, 2013 (Q)	- Dumas, 2019 - Kopytkocskiy, 2015		- Zamuda, 2013		- Dumas, 2019 - Zamuda, 2013	
	Wind Generation	- Dumas, 2019 (Q)						- Arab, 2015 - Dumas, 2019 (Q) - Zamuda, 2013						- Abi-Samra, 2011		
	Solar PV	- Aivalioti, 2015 - Dumas, 2019 (Q)	- Aivalioti, 2015 (Q) - Cotfas, 2018 (Q) - Dumas, 2019 (Q) - Zamuda, 2013													
Transmission and Distribution	Substations	- Abi-Samra, 2011 - Falabretti, 2020	- Dumas, 2019 (Q) - Sathaye, 2013 (Q)		- NGET Ada Report, 2021	- NGET Ada Report, 2021 - Sathaye, 2013	- NGET Ada Report, 2021	- Arab 2015					- Dumas, 2019 (Q) - Fant, 2020 (Q) - National Adaptation Program, 2018 (Q)	- Añel, 2017		
	Transformers	- Abi-Samra, 2010 - Añel, 2017 (Q) - Gao, 2017 (Q)	- Aivalioti, 2015 (Q) - Dumas, 2019 (Q) - Fant, 2020 (Q) - NGET Ada Report, 2021 (Q) - Sathaye, 2013 (Q)		- NGET Ada Report, 2021	- NGET Ada Report, 2021	- NGET Ada Report, 2021	- Arab 2015	- NGET Ada Report, 2021			- Fant, 2020 (Q)	- Fant, 2020 (Q)			- Fant, 2020 (Q)
	Overhead Lines	- Abedi, 2019	- Aivalioti, 2015 (Q) - Bartos, 2016 (Q) -Bigun, 2018 (Q) - Dumas, 2019 - Fant, 2020 (Q) - NGET Ada Report, 2021 (Q) - Sathaye 2013 (Q) - Zamuda, 2013		- NGET Ada Report, 2021	- Dumas, 2019 (Q) - Fant, 2020 (Q) - Sathaye, 2013 - Zainuddin, 2020	- NGET Ada Report, 2021	- Arab, 2015 - Bigun, 2018 (Q) - Dumas, 2019 (Q) - National Adaptation Program, 2018 - Ouyang, 2014 (Q)	- Fant, 2020 - NGET Ada Report, 2021		- Fant, 2020 (Q)			- Abedi, 2019 - Dumas, 2019 (Q) - NGET Ada Report, 2021		- Aivalioti, 2015 - Fant, 2020(Q) - National Adaptation Program, 2018 - NGET Ada Report, 2021
	Underground Lines	- Abi-Samra, 2010	- Aras, 2007 (Q) - NGET Ada Report (Q)		- NGET Ada Report, 2021		- NGET Ada Report, 2021				- Abi-Samra, 2010					
	Poles		- Fant, 2020 (Q) - Zamuda, 2013		- NGET Ada Report, 2021	- Dumas, 2019	- NGET Ada Report, 2021	- Abedi, 2019 - Dumas, 2019 (Q) - Ji, 2016								- Abi-Samra, 2011 - Fant, 2020 (Q)
Load Profiles	Demand	- Aivalioti, 2015 - Añel, 2017 - Bragatto, 2019 - Dindar, 2020 - Dumas, 2019 - NGET Ada Report, 2021 (Q)	- Auffhammer, 2017 (Q) - Allen, 2016 (Q) - Bartos, 2016 (Q) - De Cian, 2007 (Q) - De Cian, 2013 (Q) - De Cian, 2019 (Q) - Deschênes, 2011 (Q) - Dumas, 2019 (Q) - Ruijven, 2019 - Sathaye, 2013 (Q) - Zamuda, 2013											- Abi-Samra, 2011 - NGET Ada Report, 2021		
Upstream Fuel Processes	Fuel Storage, Terminals and Refineries										- Sathaye, 2013	- Sathaye, 2013				
	Fuel Transport					- Sathaye, 2013				- Zamuda, 2013			- Zamuda, 2013			
	Oil and Gas exploration and production									- Zamuda, 2013		- Zamuda, 2013	- Zamuda, 2013	- Añel, 2017		

Table 2.4: Key to review references.

Source	Reference number
Abedi, 2019	[39]
Abi-Samra, 2010	[40]
Abi-Samra, 2011	[41]
Aivalioti, 2015	[16]
Allen, 2016	[42]
Attia, 2015	[43]
Añel, 2017	[44]
Arab, 2015	[45]
Aras, 2007	[46]
Auffhammer, 2017	[47]
Bartos, 2016	[48]
Bigun, 2018	[49]
Bragatto, 2019	[50]
Cotfas, 2018	[51]
De Cian, 2007	[52]
De Cian, 2013	[53]
De Cian, 2019	[54]
Deschênes, 2011	[55]
Dindar, 2020	[56]
Dumas, 2019	[27]
Fant, 2020	[12]
Gao, 2017	[57]
GE, 2001	[58]
Gjorgiev, 2017	[59]
Ji, 2016	[60]
Kopytkovskiy, 2015	[61]
Linnderud, 2011	[62]
Lubega, 2018	[63]
Madani, 2013	[64]
Mehta, 2011	[65]
National Adaptation Program, 2018	[18]
NGET Ada Report, 2021	[20]
Ouyang, 2014	[66]
Qin, 2020	[67]
Ruijven, 2019	[68]
Sathaye, 2013	[69]
Vliet, 2013	[70]
Zainuddin, 2020	[71]
Zamuda, 2013	[22]

Table 2.5: Potential climate change and weather impacts on electricity grid components.

	Weather Variable / Component	Heat Waves	Ambient Temperature	Water Temperature	Temperature Amplitudes	Wildfires	Draught	Wind speed	Electric storms	Water availabilty	Precipitation	Sea level Rise	Flood	Cold Waves / Ice storms	Snowfall	Vegetation
Gneration Technologies	Thermoelectric Generation - Gas	- Efficiency reduction and capacity reduction due to high coolant temperature. - Reduced power output due to lower fuel mass ignition.	- Efficiency reduction and capacity reduction due to high coolant temperature. - Reduced power output due to lower fuel mass ignition.	- Efficiency reduction and capacity reduction due to high coolant temperature. - Reduced power output due to lower fuel mass ignition.			Curtailed production due to reduced water availabilty. Possibilty of water level dropping below intake valves.	Risk of physical damage.		- Curtailed production due to reduced water availabilty. Possibilty of water level dropping below intake valves. - Supply chain risks.	- Fuel supply risks. - Flooding risks. - Risk of damage due to land movements.	Risk of physical damage for coastal facilities.	Risk of physical damage for in-land and coastal facilities.	Reduced fuel delivery due to iced pipe systems		
	Thermoelectric Generation - Nuclear	- Efficiency reduction and capacity reduction due to high coolant temperature. - Safety concerts due to elevated temperatures around emergency diesel generators.	Efficiency reduction and capacity reduction due to high coolant temperature	- Efficiency reduction and capacity reduction due to high coolant temperature. - Curtailment due to esign and regulatory limits reached for intake and outake water temperatures			Curtailed production due to reduced water availabilty. Possibilty of water level dropping below intake valves.	Risk of physical damage.		- Curtailed production due to reducedcooling potential. Possibilty of water level dropping below intake valves. - Supply chain risks.	- Fuel supply risks. - Flooding risks. - Risk of damage due to land movements.	Risk of physical damage for coastal facilities.	Risk of physical damage for in-land and coastal facilities.			
	Hydroelectric Generation	- Capacity decrease due to higher water evaportation. - Operation changes.	- Capacity decrease due to higher water evaportation. - Operation changes.				Generation decrease due to perturbed annual runoff cycles or air-bubble formation.			- Generation decrease due to perturbed annual runoff cycles or air-bubble formation.	- Physical damage risks due to excessive runoff and land movements. - Generation decrease due to perturbed annual runoff cycles.		- Risk of physical damage. - Operation changes		Generation decrease due to perturbed annual runoff cycles.	
	Wind Generation							- Planning risks due to uncertain wind patterns. -Risk of physical damage (loss of blade and support tower buckling) due to wind speeds exceeding design limit.						Risk of physical damage.		
	Solar PV	- Electrical Efficiency and power output reduction	- Electrical Efficiency and power output reduction													
Transmission and Distribution	Substations	Derating, life-loss and failure risks.	Derating, life-loss and failure risks.		Component fatigue.	Risk of physical damage.	Risk of physical damage due to ground movements.	Risk of physical damage, especially due to falling trees.					Risk of physical damage.	Higher breakage probabilty due to ice accretion.		
	Transformers	- Derating, life-loss and failure risks. - Reduction in efficiency and maximum capacity due to risk of exceeding design hot spot temperature and triggering paper insulation combustion (total failure)	- Derating, life-loss and failure risks. - Reduction in efficiency and maximum capacity due to risk of exceeding design hot spot temperature and triggering paper insulation combustion (total failure)		Component fatigue.	Risk of physical damage.	Risk of physical damage due to ground movements.	Risk of physical damage, especially due to falling trees.	Risk of physical damage.			Risk of physical damage.	Risk of physical damage.			
	Overhead Lines	- Derating, life-loss and failure risks. - Increased resistive losses. - Derating to protect from loss-of-life. - Sagging risks due to conduction expansion.	- Derating, life-loss and failure risks. - Increased resistive losses. - Derating to protect from loss-of-life. - Sagging risks due to conduction expansion.		Component fatigue.	- Risk of physical damage. - Higher electrical arcing risks due to ash levels in the air. - Sagging risks due to conduction expansion.	Risk of physical damage due to ground movements.	Risk of physical damage, especially due to falling trees.	Risk of physical damage.		Higher breakage probabilty if combined with low temperatures due to ice accretion.			Higher breakage probabilty due to ice accretion.		Increased losses and failure risk due to tree growth.
	Underground Lines										Corrosion risks.	Corrosion risks.				
	Poles		Increased rupture risks due to moulding.			Risk of physical damage.	Risk of physical damage.	Risk of physical damage, especially due to falling trees.								Increased rupture risks due to moulding.
Load Profiles	Demand	- Increased cooling demand. - Reserve margin risks.	Increased cooling demand.											Increased heating demand.		
Upstream Fuel Processes	Fuel Storage, Terminals and Refineries										Productions risks due to lower water availabilty.	Risk of physical damage.				
	Fuel Transport					Risk of physical damage and interrupted operations.				Reduced fluvial transport potentials due to lower river levels.			Risk of physical damage and interrupted operations due to flooded roads and rails.			
	Oil and Gas exploration and production	Damaged infrastructure due to ice melt.								Lower fracking and exploration potentials due to lacking water resource.		Risk of physical damage.	Risk of physical damage.	Risk of physical damage.		

Chapter 3

Data Collection and Analysis Methodology

3.1 Data extraction and quantification selection

This section summarises the necessary data for the impact assessment carried out by this thesis, the associated litterature sources and the analysis methodology. It details the selection process for the sources used in this work's review (Section 3.1.1), their classification (Section 3.1.2), and the extraction of the reported quantifications in Table 3.3. Choices of the studied weather variables and used correlations are detailed in Section 3.1.4 and Section 3.1.5, respectively. Finally, the climate data selection is outlined in Section 3.1.6. All data inputs, sources and selection criteria are summarised in Table 3.1.

Table 3.1: Summary of extracted data, literature sources and criteria.

Data type	Literature source	Criteria
References	Scientific and industry literature	Keywords - Published 2012 and later
Quantifications	Review	Equations and linear relationships
Impact functions	Quantifications	Linear relationships - Clear reference temperatures
Climate data	UKCP18 Global (60 km) and Local (12 (km)	Max daily and monthly data Avg and Max ambient temperature - for 2030-2050 and 2080-2099

3.1.1 Methodology for the review of climate-impact studies on electricity systems

In order to study the pressures of Weather Variables (WV) on GCs, it was first necessary to establish a state-of-the-art review of works reporting on those topics. Published scientific articles and industry reports were collected using combinations of keywords relevant to electricity network and climate studies. These are summarised in Table 3.2

A loose publication date limit was set at 2012, in order to fit this review in the thesis' available time frame. This date coincides with the year of the "Climate Change and Extreme Weather Vulnerability Assessment of the US Energy Sector", which led to the production of the first studied wide-scale impact review [22]. Limiting the publication dates also allowed accounting for the evolutions in power technology and infrastructure standards, bringing studied components closer to current assets.

Table 3.2: Reference-search keywords by category.

Weather related	Network related	Impact related
Extreme weather	*Grid component*	*Impact*
Heat waves	*Infrastructure*	*Failure*
Air temperature	*Power*	*Vulnerability*
Ambient temperature	*Transformer*	*Recovery*
	Transmission lines	*Resilience*
	Solar PV	*Losses*
	Demand response	*Risk factor*
	Hydroelectricity	*Temperature coefficient*
	Nuclear power	

3.1.2 Litterature sources classification

The collected references were classified in a table according to the mentioned WVs and GCs. This provided a landscape of the various ways that the weather variables can influence the operations of grid components. Heatwaves, ambient-temperature levels and temperature amplitudes could be similar, but they were separated because literature commented on the specific impacts of each one of them separately from the others. Heatwave events were usually defined with temperatures above a given threshold for a set number of consecutive days. In contrast, references using ambient temperature usually looked at hourly or daily values regardless of "extreme" events. Vegetation was included as a WV although its interference with assets is not a direct impact of meteorological events (tree fall on assets is related to wind and thus classified under that WV). Some sources argued that climate change affects vegetation growth, increasing risks to transmission assets like overhead lines [12]. Other sources, mentioning humidity-related land movements, were classified under Drought or Precipitation depending on the paper's context, as those are the primary variables driving the subsequent impact on assets.

3.1.3 Identifying quantifications

In order to proceed with a UK-wide impact analysis, it was necessary to identify relationships between WV values and impact levels on GC. Those were usually provided as iteratively-solved equations or linear relationships. A (Q) sign identifies the references falling into that category in the summary Table 2.3. Computational modelling and impact quantifications made for particular warming levels – like +2 °C by the end of the century – were not marked with (Q) since they do not provide an explicit correlation between variable observation and impact.

3.1.4 Selecting a weather variable for further study

The focus of this thesis on ambient temperature, as a unique weather pressure variable, was decided based on considerations related to the time-frame and computational resources available to this thesis. The quantity of references studying temperature impacts suggests that they constitute a significant risk for ESs. The National Grid Electricity Transmission (NGET) Climate Adaptation Report places ambient-temperature effects on transformers, lines, switchgear and demand at the highest levels of their risk assessment in 2050 [20]. Ambient temperature was therefore chosen to establish the proof-of-concept for impact visualisations such as the one developed here. Additionally, the number of available quantified relationships enables impacts to be evaluated with better contextual accuracy.

3.1.5 Impact functions and final correlations selection

In order to successfully calculate impact ranges from the selected WV, component-specific and linear impact functions were extracted from the identified set of quantifications. A broad assessment was favoured over individual GC impact precision. Final correlations were therefore selected according to their applicability in the UK context and ease of implementation in later modelling. Temperature-related damage functions on GCs usually display linear relationships [69]. The ones

finally extracted rely on reference temperatures between 15°C and 25°C, which is a standard operating range for powerplants and transmission assets. Additionally, it was deemed safe to assume that linear relationships were appropriate for current and for future temperature ranges. This is justified because the shape of impact functions is only likely to change for temperatures higher than those expected in the future [69].

In accordance with the exploitable found correlations, components selected for impact analysis were: gas generation, nuclear plants, solar PV installations, transmission transformers, overhead lines, and power demand. Table 3.3 summarises the selected correlations, and details of the selection process are given in the paragraphs below.

Table 3.3: Impact functions per analysed grid component. All equations are expressed in % reduction of rated values, where P is the component's power output, C_{max} is the component's maximum capacity, I_{max} is the component's maximum permissible current and L_{peak} is the resulting daily peak load.

Grid Component	Variable	Equation	Threshold (T_{thresh}) in ºC
Gas	Max temperature	$P = 0.7T_{amb} - 11.37$	16
Nuclear	Max temperature	$P = 1.03(T_{amb} - T_{thresh})$	20
SolarPV	Max temperature	$P = 0.48(T_{amb} - T_{thresh})$	25
Transformers	Max temperature	$C_{max} = 1(T_{amb} - T_{thresh})$	20
Overhead Lines	Max temperature	$I_{max} = 1.3T_{amb} - 30$	23
Demand	Avg temperature	$L_{peak} = 3.7(T_{amb} - T_{thresh})$	15

Thermoelectric Generation - Gas

Combined Cycle Gas Turbines (CCGT) plants in the UK use three cooling strategies: a) they rely on water circulating systems (once-through or recirculating), b) they use air for cooling or c) require no cooling at all. Each method comes with different sensitivities to ambient temperature. In 2014, 82% of thermoelectric plants in the UK used water for cooling [72]. Cooling-method choices are usually strongly influenced by local meteorological conditions. Details were not provided for CCGT, but, if similar proportions apply, most plants use wet cooling systems. It is important to note that those rely on available water resources sensitive to drought and heatwave events (not studied in this work). Although elevated air temperatures more heavily impact dry-cooled systems, these could prove more resilient in case of water stress.

Figure 3.1 presents linear relationships of percent loss in rated power-output capacity and ambient temperature of Combined Cycle Gas Turbine plants. These were extracted from the literature review, for different studies, and are represented by the following labels: GE (turbine data sheet), Dumas Wet Low, Dumas Wet High, Sathaye Dry, Sathaye Wet Valley. The General Electrics (GE) line is based on the "General Electric turbine data sheet", where the results of lab testing of a turbine in isolation are shown [73]. The "Dumas Wet Low" and "Dumas Wet High" correspond to wet-cooled CCGT responses at low- and high-impact limits, respectively [27]. The "Sathaye Wet Valley" line is based on data from a study reporting temperature-related average impacts on valley-located water-cooled plants in California. These operate in relatively similar meteorological conditions to those in the UK. Finally, the "Sathaye Dry" line represents data of a study reporting average impacts on air-cooled turbines [69].

Any of the presented relationships were considered suitable for further impact assessment. Overall, rated-capacity reductions per degree of ambient temperature increase above 15 °C ranged from 0.26% to 0.76%. The Sathaye Wet Valley correlation was discarded due to the impact being non-zero at the reference design temperature (15 °C). Sathaye Dry was finally used for later calculations as dry-cooled gas plants are more sensitive to ambient-temperature changes [74]. Additionally, the correlation drew on data from plants in meteorologically similar conditions to those found in the UK, allowing for an assessment of a realistic 'worst-case' scenario.

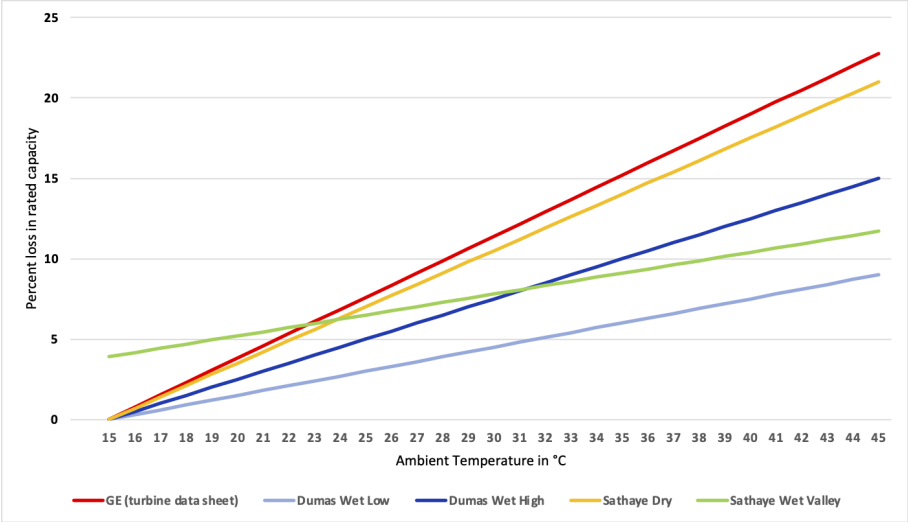


Figure 3.1: Percent loss in rated capacity (power output) with ambient temperature of Combined Cycle Gas Turbine plants across studies.

Thermoelectric Generation - Nuclear

Climate-impact studies on nuclear generation plants usually rely on water-temperature observations [27]. The only study relating ambient-temperature levels to nuclear-plant performance, found in the review perfomed for this thesis, uses a mixed methodology for impact-functions definition, described in Section 2.4.1 [62]. This assement relies on regression techniques over empirical datasets to determine analytically-defined regressors. Two approaches allowed controlling those regressors. The first involved a plant-specific approach based on empirical operational data from a boiling water 1,200 MW reactor. The second relied on a panel-data approach using data from thermal power plants (mixed technologies) across seven European countries [62]. Thermal-efficiency changes, alone, are shown to decrease power output by 0.3-0.4% per degree rise in ambient temperature between -7 °C and 20 °C, and around 0.96-1.10% above 20 °C. It is also suggested that factoring in cooling-water availability and regulatory limitations on rejected water temperature could lead to losses exceeding 2% per degree.

Regarding this work's impact estimations, the retained relationship is an average value of 1.03% per degree above 20 °C. The sensitivity factor is higher than the one found for gas-powered plants. This is consistent with the added vulnerabilities of nuclear generation, i.e. lower generation efficiency, higher water requirements and lower cooling efficiency, all of them impacted by air temperatures [62].

Solar Photovoltaic

Reported temperature coefficients show a reduction in maximum power output of 0.48% per degree for monocrystalline silicon (mSI) and 0.38% per degree for polycrystalline silicon (pSi) cells, for conditions above 25°C and at solar irradiation of 1,000 W/m^2 (standard testing conditions). The relationship for mSi was used for impact calculations as this technology is more common commercially and more sensitive to temperature changes [75]. Additional sources [16, 27] confirmed that efficiency losses between 0.4% and 0.5% lie within expected ranges.

Transformers

The reported correlations, linking ambient temperature to impact on transformers, show strong variations in sensitivity factors. One study indicated that a transformer's maximum capacity decreases by 0.7% per degree above 30 °C [69]. A second study suggested a 1% decrease per degree above 20 °C [16]. Reviews like [27] report factors in both ranges. The difference cannot be explained by varying sensitivities between lower-voltage distribution transformers and higher-voltage transmission, as they were shown to display close to identical sensitivities [76]. The found variations could be the result of some works studying operating conditions to failure (i.e. working temprature range before combustion of the internal paper insulation). In contrast, other works calculate the maximum no loss-of-life capacity (i.e. a derating of loading values that would allow operations at those temperatures without loss-of-life). This second approach would lead to a more conservative result. The relationship with the highest sensitivity factor was finally selected for further analysis, presumably simulating the required derating of UK transformers in no loss-of-life operation modes. Nevertheless, one should note that the lower factor could allow simulating operations during short-lived high-temperature extremes, assuming components would be permitted to operate beyond their limits during high grid instability. As the UK regulation and planning for such operations were not studied, it was judged safer to carry on the analysis with the higher factor.

Overhead Lines

The reviewed works typically study ambient-temperature impact on overhead transmission lines, assuming standard ambient conditions of wind speed and solar irradiation. These variables affect the line's operating temperature and, therefore, its resistive losses and its life expectancy. Depending on the used conductor, lines have a design limit for operating temperature between 75°C [48] and 80°C [69]. The direct impact of increased ambient temperature on a conductor's resistance leads only to low capacity losses [69]. As reported by one study, the IEEE's technical parameters for a sample line show only a 0.14% capacity change between 38°C and 43°C. It is unlikely that resistive losses in transmission lines will lead to significant impacts by the end of the century. However, a line's carrying capacity is usually lowered by grid operators when their operating temperature is in danger of exceeding the line's design limit, as higher operating temperatures could lead to line damage. Induced capacity losses (also called line derating) should be considered, assuming that grid operators will keep avoiding line loss-of-life.

Line derating is estimated to lead to losses of a similar range across various sources: 1.5% loss per degree between 38°C and 43°C of ambient temperature [69], or 1.3% loss per degree for temperatures between 30°C and 40°C [49]. This second correlation was chosen because its starting threshold temperature is lower, and the relationship was extended to include higher temperatures. For reference, it was estimated that US transmission lines could lose an average of 2-6% of their carrying capacity in the summer season due to capacity derating [48].

Peak Demand

Energy demand is influenced by ambient-temperature levels [27, 22]. However, it is primarily a function of population numbers and income levels [54], seasonality [77], geography [47] and fuel prices [54]. To control for population levels, impacts were measured per capita or using spring average demand levels as a reference. Temperature variations can have short- and long-term influences on demand [77]. The subsequent calculations do not account for future adaptations in behaviour or technology to the rising temperatures.

As a general trend, energy use falls with rising temperature because of reduced heating needs, but it can simultaneously increase due to additional cooling demand [77]. The first effect mostly impacts gas and oil use as they are the primary heating fuels. On the other hand, the increase in demand for cooling primarily impacts electricity use as it is usually driven by Air Conditioning (AC) use and household appliances. Income levels and fuel prices have also been shown to influence energy use, but mainly in the long term rather than the short-term ([54]. As this study focuses

on the short term impacts of high temperatures, wealth levels and fuel prices were not projected along with the ambient temperature. Peak load has been judged as more relevant than average demand to understand the demand pressures that could bring grid components to fail. Short-term demand increases are more likely to bring power generation and transmission capacities to their limits as variations are sharp and more challenging to predict.

Sensitivities show substantial variations across works. One study projects a decrease in UK residential electricity demand by the end of the century and a sharp increase in respective demand for the US over the same period [77]. It should be noted here that studies incorporating commercial and industrial energy use could have very different results. These consumers have higher sensitivities to ambient temperature than the residential sector in temperate-climate countries [54]. When looking at annual energy consumption, it is estimated that annual residential energy consumption in the US could increase by 11% in a high-emissions scenario [55]. Regarding peak-load studies, one source provides a state-wide peak-load function in Watts per capita [69]. Linear regression shows an increase in demand of around 4% per degree above 24°C. A different study focuses on peak demand and its response to temperature across US networks [47]. The response of the southeast US network PJM was extracted to establish a general correlation between daily peak demand and average daily temperature. This particular network was chosen as it is one of the largest competitive wholesale markets in the world and is representative of "colder" regions, with heating and cooling patterns relatively close to those in the UK. Electrification of heating and Air Conditioning (AC) penetration are assumed to be similar in the states covered by the PJM network and in the UK. The response graphs were provided in MW of extra demand compared to the spring average demand value. This reference value was not provided but taken to be 1,200 MW for that network from a separate source [78]. The obtained sensitivity factor was a 3.7% increase in peak load per degree above 15°C. This value is consistent with other networks studied by [48].

Finally, the linear relationship computed for the largest US network, categorised as representative of cooler regions, was chosen [47], while being aware that resulting impacts will be higher than those obtained with other correlations.

3.1.6 Climate data selection

This work's climate data originates from the UKCP data bank. The chosen scenario was RCP8.5, simulated under UKCP18 model sets Global (60 km) and Local (2.2 km - regridded to 5 km). This specific scenario has been used with UKCP18 models of different geographical resolutions, which opens the possibility of future iterations of this analysis at increased resolutions. It also represents a useful "worst-case" scenario for climate-resilience assessments.

Average and maximum temperatures were recovered as daily and monthly data for the 60 km and 5 km scales over two 20-year periods - from 2030 to 2050 and 2080 to 2099. Data from 28 members were available at 60 km scale – 15 from PPE-15 and 13 from CMIP5-13 – while 12 members were available at the 5 km scale.

3.2 Tool production, data treatment and result display

3.2.1 Building an impact calculation program

One of the goals of this thesis was to create a versatile and easy-to-use program for GC climate impact calculation. The program needed to provide the flexibility to load climate data with varying time and space resolutions over custom periods. The intention was to provide users with a tool that they could easily use to choose the components and weather variables to study, as well as impact functions. These criteria were chosen as design requirements to obtain the program's high adaptability and growth-oriented structure. Although only two variables (maximum and average daily temperatures) were loaded in this version, this program's structure allows quick incorporation of other weather parameters and new GCs into the calculation process.

A custom Object-Oriented program was developed. It ts powered by four modules illustrated in Fig. 3.2: the Main Script, a Data Object Class, a Grid Component Class and a Map Class. The program modules' broad functionalities are: initialising the variables under study, seting impact functions and thresholds and calculating impact levels. The calculation processes, mentioned and illustrated in Fig. 3.3, are described hereafter. The main script module runs ten consecutive segments. Initially, segments 1 and 2 load necessary packages and data paths. Segment 3 defines default GC and WV sets, as well as their correlations and thresholds. The default elements are those described in Section 3.1.5. If users desire to study different GC or WV, or want to modify default correlations, this is where they can do so. In segment 4, Data Objects (DOs) are initialised, enabling climate data loading. Each DO represents one studied WV and allows loading and treating one dataset. Users can add WVs to this section to initialise custom WVs datasets. Segment 5 prompts the data pre-treatment described in Section 3.2.2 using the initialised DOs. Segment 6 generates the WV maps for result visualisation. Segment 7 initialises Grid Component Objects (GCOs), giving to each one a separate identification and preparing them for impact calculation. This is where changes to GC sets can be implemented if desired. Segment 8 prompts impact calculations, while segments 9 and 10 ensure results are saved in csv files and geojson map files.

3.2.2 Climate data pre-treatment

In order to obtain data representing high ambient-temperature extremes or "extreme hot days" with a satisfactory degree of likelihood, a shedding of values, judged as too "extreme", was done on the UKCP18 datasets. First, a Kolmogorov-Smirnov test was run with the parameters summarised in Table 3.4 across member data [79]. This established that data for each time-space point followed a normal distribution with a confidence level of α=0.05. The p90 value was then selected across members for each time-space point (where p90 denotes the value for which 90% of observations fall below that value). There is, therefore, a 90% probability of observations falling below the recovered p90 values, and the top 10% of most "extreme" ambient temperature values will be excluded. The absolute maximum across the period of interest was recovered to draw result maps in Section 4.3.2. These show the temperature values used for subsequent impact calculations for the 2030-2050 period and the 2080-2099 period respectively.

Table 3.4: Parameters for normal distribution Kolmogorov-Smirnov test across member data.

Test element	Description
Null hypothesis	H_0 : data is normally distributed across members
Alternative hypothesis	Ha : data is not normally distributed
Significance level	$\alpha = 0.05$
Critical region	Reject H_0 if $D > D_crit$
Critical value	$Dcrit = 0.25$ for 28 members
	$Dcrit = 0.375$ for 12 members
Test statistic	D

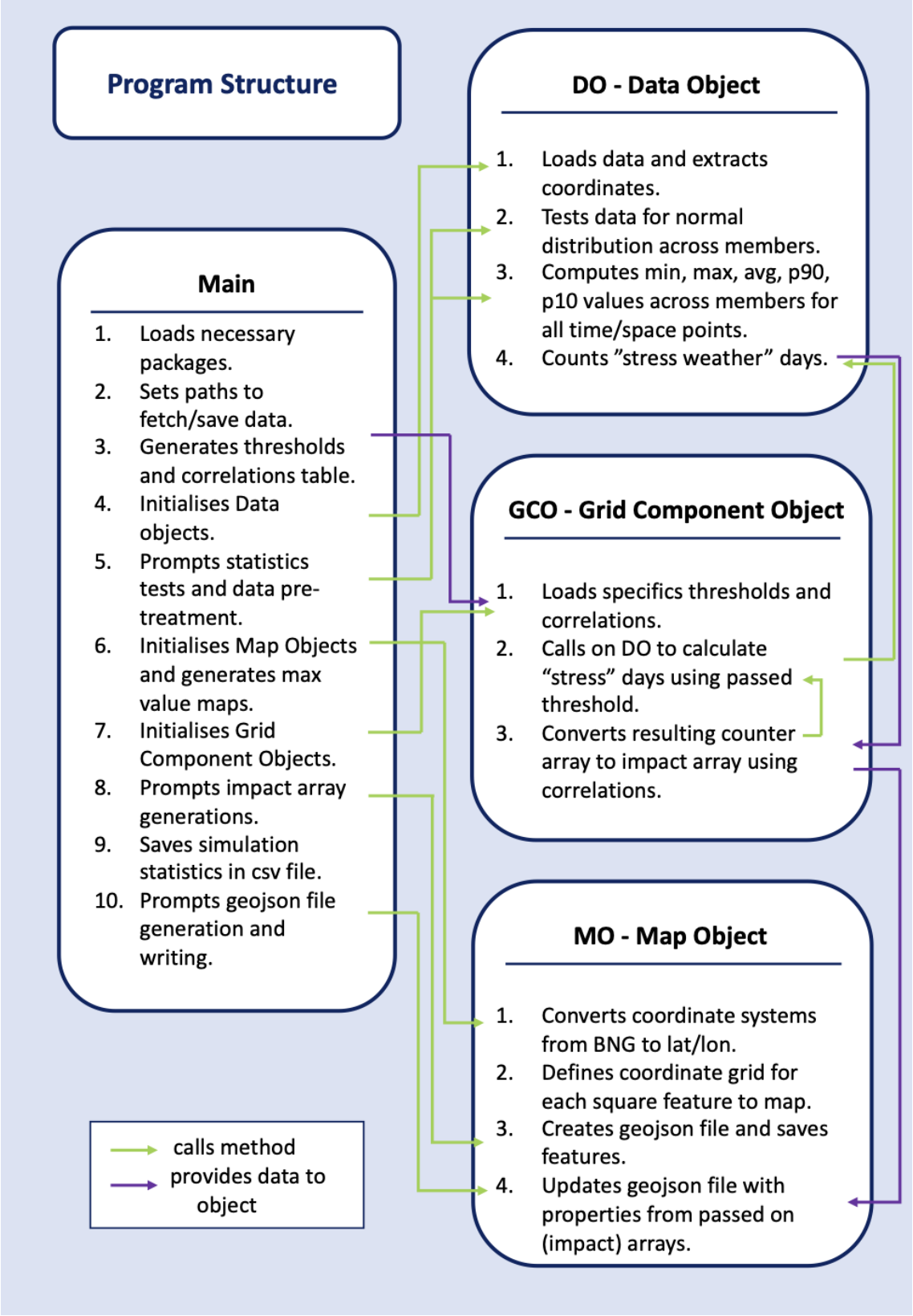


Figure 3.2: Program structure for the calculation of climate impacts on electricity grid components.

3.2.3 Calculating impact

The produced datasets allowed obtaining maximum-impact values for GCs over two 20-year periods (2030-2050 and 2080-2099) at every geographical grid cell. For each studied GC, there was first a selection of "stress-days" where temperatures exceeded the relevant threshold. The correlations defined in Table 3.3 were applied to those "stress-days" in order to obtain impact levels from p90 maximum-temperature days. Maximum values of impact were recovered across time for each spatial grid cell. The time-wise maximum of those impact values was computed for each spatial grid cell and displayed in the result maps of Section 4.3.2. The associated impact statistics, presented in Section 4.3.1, were produced by extracting the maximum, minimum and average impact values across the daily maximum impacts mapped in each cell. All the described data-treatment steps are illustrated in Fig 3.3.

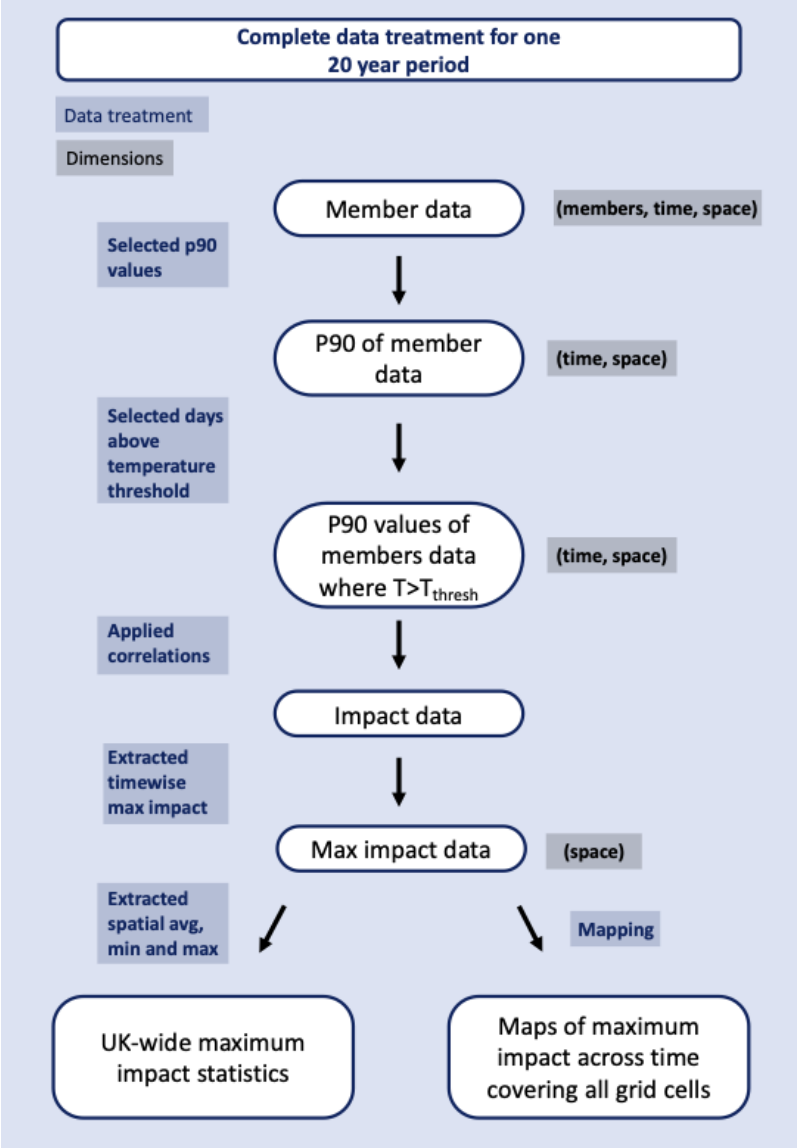


Figure 3.3: Complete data treatment for the evaluation of climate impacts on one grid component over a 20-year period.

3.2.4 Mapping and visualisation tool development

This work aimed to deliver a proof-of-concept for easy-to-grasp visualisations of climate impacts on electricity grids. A web-based interface was chosen for its ease of access and dissemination potential, allowing users from the wider public to access it and interact with the data. The visualisations obtained using this interface are described in the following Chapter.

Chapter 4

Results, Discussion and Conclusion

4.1 The Grid Climate Resilience Tool

The most important result of this thesis work is the delivery of a program that can be tailored to provide impact quantifications on custom grid-component and weather-variable sets. The calculation program described in Section 3.2.1 was named the Power Grid Climate Resilience Tool (GCRT). It uses Python 3 as a programming language [80] and the version tracking was done using GitHub [81]. The complete program is openly accessible and hosted at the following link: `https://github.com/Kfsand/gclimateres`, allowing the code to be consulted, customised, and upgraded.

Its folders include default raw datasets and files used to generate the results described in Section 4.3.1 and Section 4.3.2 below. The program requires the following inputs to run an impact simulation: 1) climate data in csv format, with values provided for grid cells over a particular geographical area and period; 2) grid components and weather variables, with specified correlations and thresholds for the impact calculations; 3) input and output resolutions and 4) desired parameters for data testing and pre-treatment.

4.2 Visualisation web application

The developed web application for the results visualisation can be found in a custom-made website, hosted on GitHub Pages at the following link: `https://kfsand.github.io/gclimateres/`. The used maps are powered by the MapBox Studio API and were generated using the geojson files provided by the GCRT. Its interface allows users to view maximum temperature maps as default, or select a specific grid component and view impact levels for the 2030-2050 and 2080-2099 periods.

4.3 Simulation results

4.3.1 Impact summary

The maximum daily impact values, obtained across the UK over the 2030-2050 and 2080-2099 periods, are summarised in Table 4.1 and Table 4.2, respectively. By mid-century, the technology losing the most considerable portion of its power generation capacity, on maximum impact days, is gas, with an average loss across the UK grid cells of 13.08%. Nuclear power follows closely at a loss of capacity of 12.78%. The order inverts when looking at the grid cell with the highest impact for each one, with nuclear power losing 18.77% of its capacity at the most and gas 16.86%. Solar PV is much less affected, with an average national-wide loss of 3.56%, relative to rated capacity, and a maximum loss of 6.35%. Transmission assets, like transformers, lose 12.41% of their rated load capacity on average and 18.22% at most. Meanwhile, overhead lines experience an average decrease in current-carrying capacity of 12.13% relative to rated and a maximum decrease of 19.69%. Demand is the studied component with the strongest sensitivity to ambient temperature, with the UK-average increase in daily peak load reaching 38.96% and 53.64% as a maximum.

All the described metrics reach higher values for 2080-2099 than for 2030-2050. Over that period, gas generation loses 18.31% capacity on UK average (max 22.78%). Nuclear generation loses 18.31% capacity on average across the UK (max 25.76%). Solar PV generation loses 6.78% capacity on UK average (max 9.61%). Transformers see their load capacity reduced by 19.12% on UK average (max 25.01%), while overhead lines have their carrying capacity reduced by 20.86% on UK average (max 28.52%). The component affected the most is demand once again. The average peak load increases by 57.05%, and an absolute maximum is reached at 70.85%.

Table 4.1: Geographical statistics on UK-wide maximum ambient-temperature impacts on selected grid components - 2030-2050 period (where Min is the minimum, Max is the maximum and Avg is the average of the highest ambient-temperature impacts across the UK, respectively)

Grid component	Min	Max	Avg	Impact metric
Gas	9.28	16.85	13.08	% loss in generation capacity relative to rated
Nuclear	7.44	18.77	12.78	% loss in generation capacity relative to rated
Solar PV	1.07	6.35	3.56	% loss in generation capacity relative to rated
Transformers	7.33	18.22	12.41	% loss in load capacity relative to rated
Overhead Lines	5.39	19.69	12.13	% loss in carrying capacity relative to rated
Demand	22.6	53.64	38.96	% increase in daily peak load relative to spring average

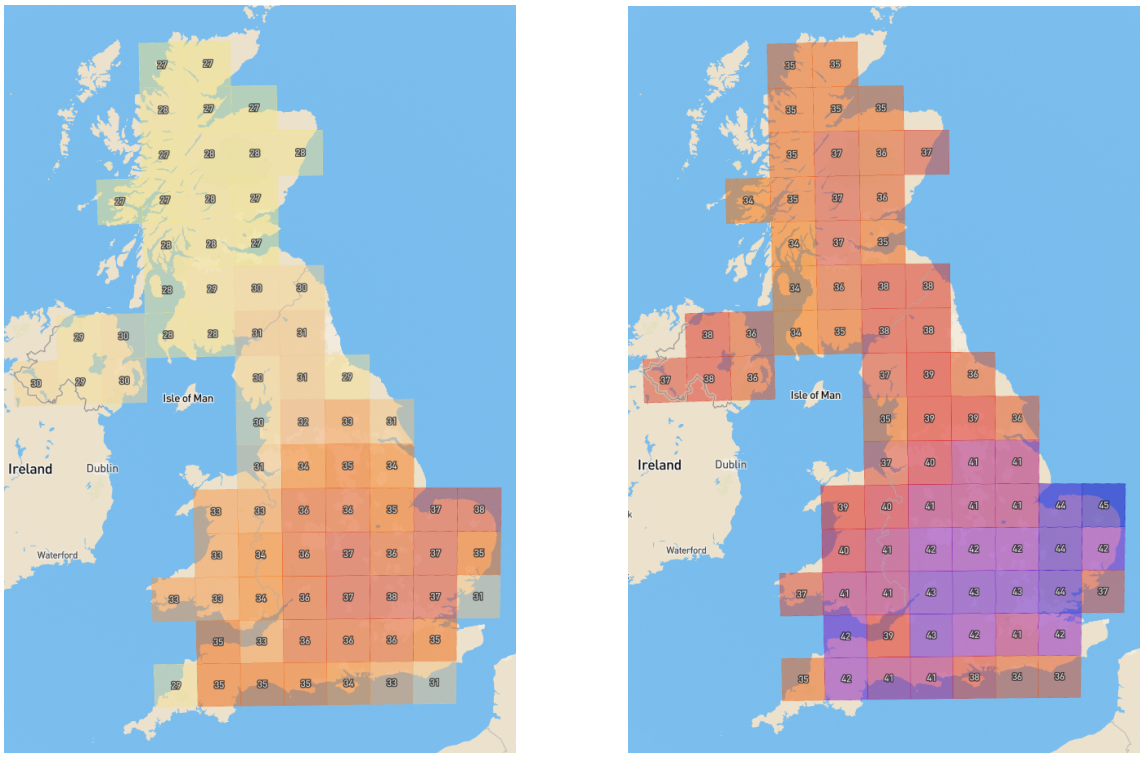
Table 4.2: Geographical statistics on UK-wide maximum ambient-temperature impacts on selected grid components - 2080-2099 period - (where Min is the minimum, Max is the maximum and Avg is the average of the highest ambient-temperature impacts across the UK, respectively)

Grid component	Min	Max	Avg	Impact metric
Gas	14.63	22.78	18.31	% loss in generation capacity relative to rated
Nuclear	14.7	25.76	18.31	% loss in generation capacity relative to rated
Solar PV	4.45	9.61	6.78	% loss in generation capacity relative to rated
Transformers	14.27	25.01	19.12	% loss in load capacity relative to rated
Overhead Lines	14.55	28.52	20.86	% loss in carrying capacity relative to rated
Demand	43.93	70.58	57.05	% increase in daily peak load relative to spring average

4.3.2 Impact maps

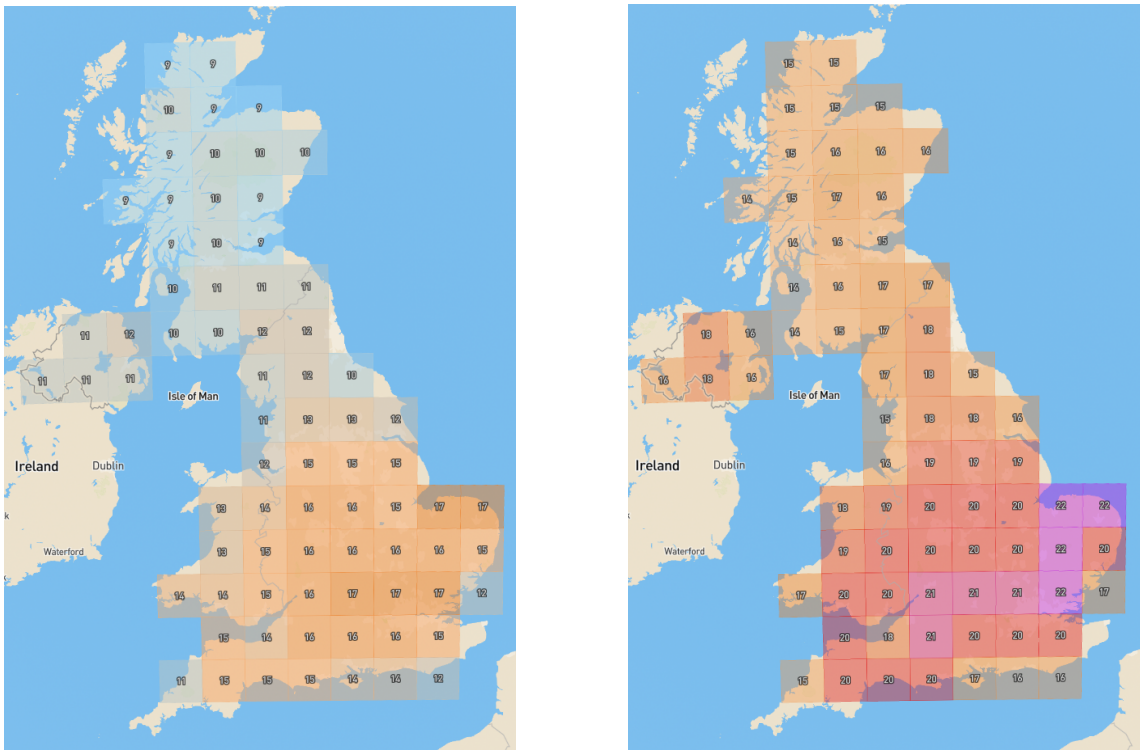
The processed temperature data displayed in Fig. 4.1 show that maximum temperature values tend to be the highest in the southeast of England and Wales. The South-East, East Anglia, and East-Midlands regions display the highest temperatures, consistently over 30°C (38°C. max) over 2030-2050. Meanwhile, Scotland and Northern Ireland remain much cooler, with temperatures reaching barely over 30°C. A similar pattern repeats over the 2080-2099 period. The South-East, East Anglia, and East-Midlands regions consistently reach temperatures between 40°C and 45°C. Scotland and Northern Ireland remain under the 40°C mark, with temperatures spanning 35°C to 38°C on maximum.

The impact maps produced for gas generation in Fig. 4.2, nuclear power in Fig. 4.3, solar generation in Fig. 4.4, transformer capacity in Fig. 4.5, overhead line carrying capacity in Fig. 4.6, and daily peak demand in Fig. 4.7, show patterns similar with those of temperature. The impact values obtained for 2080-2099 are all consistently higher than those obtained for 2030-2050. The highest impact values are reached in the southeast of England, while Scotland and Northern Ireland generally suffer a lower impact.



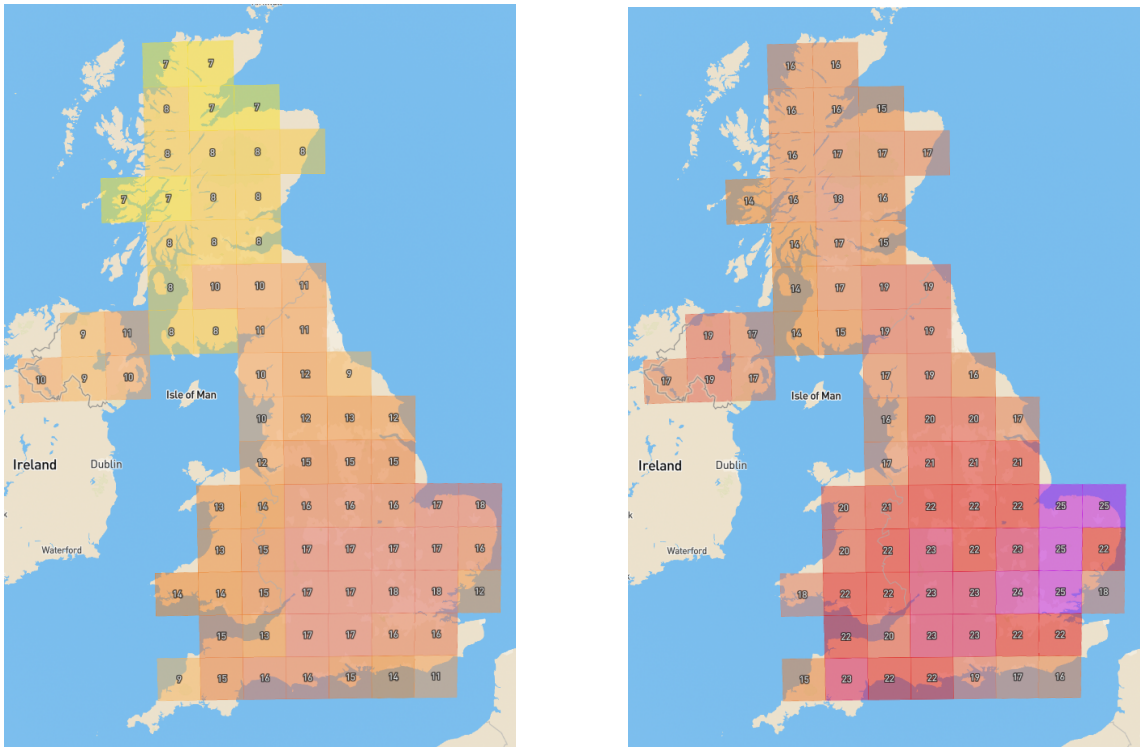
(a) 2030 – 2050 (b) 2080 – 2099

Figure 4.1: Maximum daily temperature (°C) reached in the UK across two 20-year periods - (60 km^2 resolution)



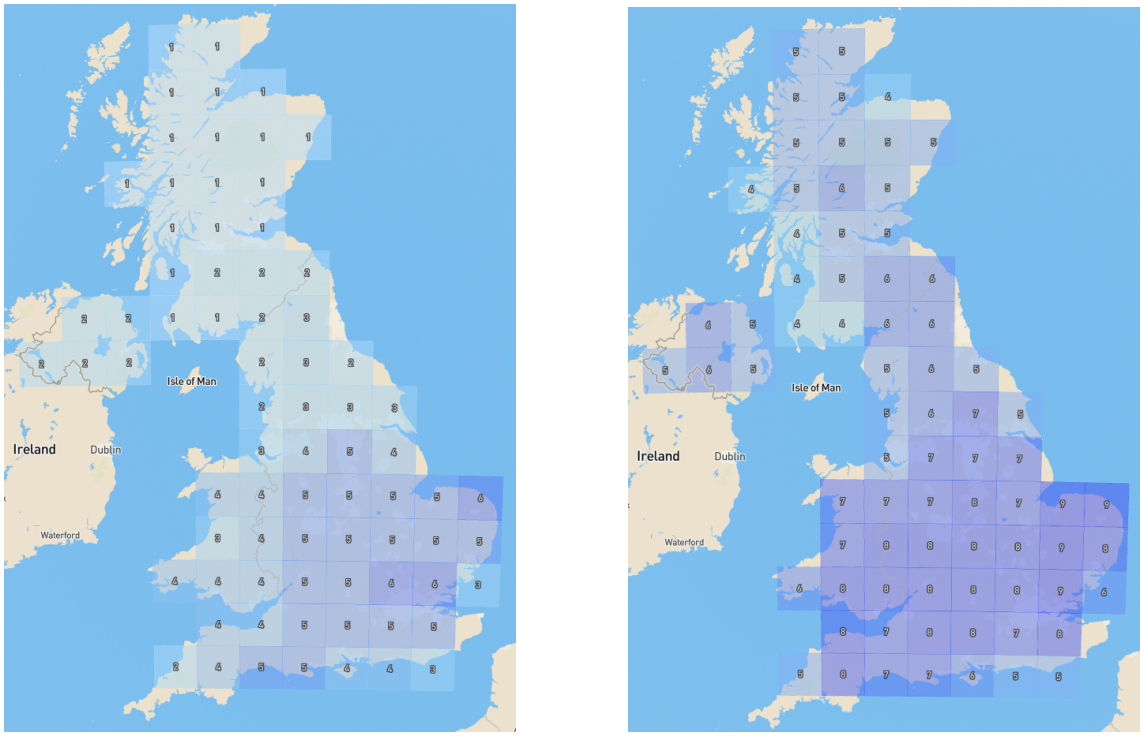
(a) 2030 – 2050 (b) 2080 – 2099

Figure 4.2: Maximum loss of generation capacity for Gas plants (% of rated) in the UK - (60 km^2 resolution)



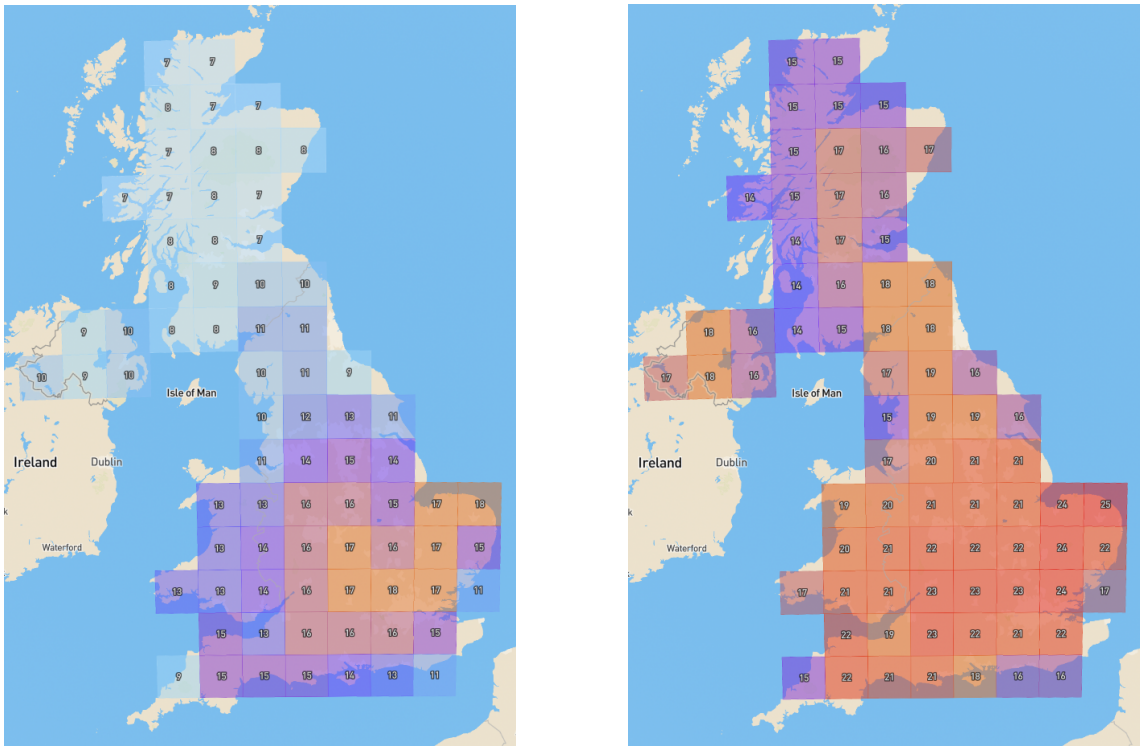
(a) 2030 – 2050 (b) 2080 – 2099

Figure 4.3: Maximum loss of generation capacity for Nuclear plants (% of rated) in the UK - (60 km^2 resolution)



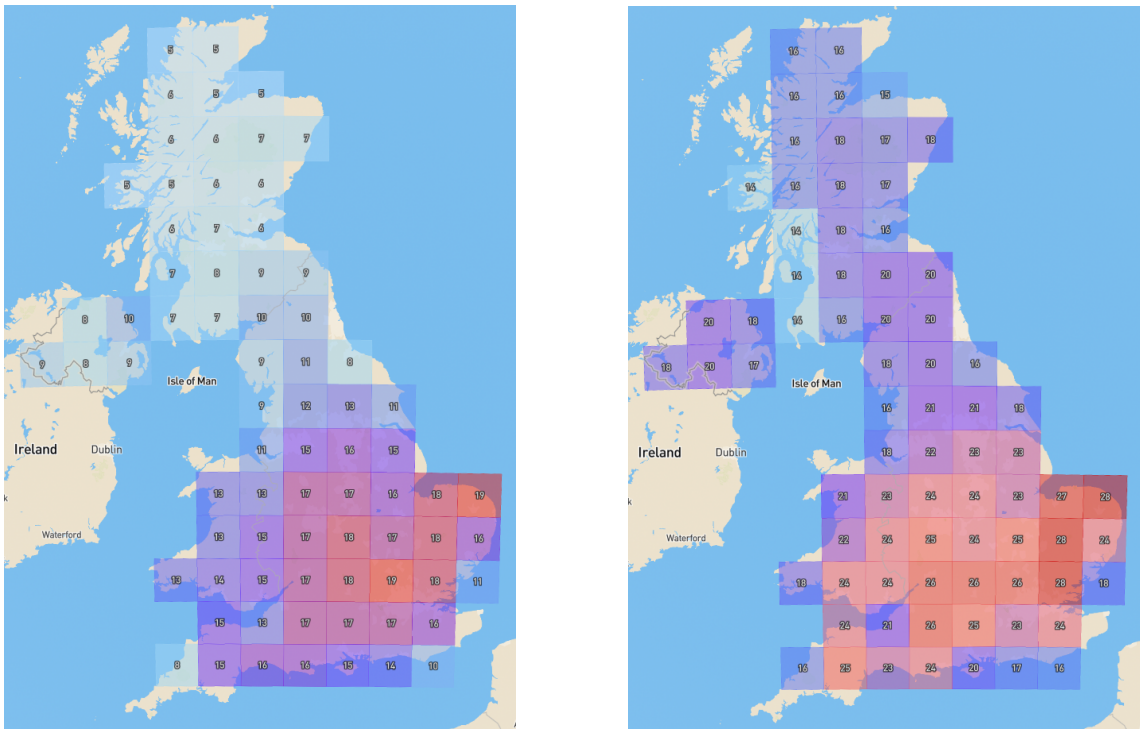
(a) 2030 – 2050 (b) 2080 – 2099

Figure 4.4: Maximum loss of generation capacity for Solar PV (% of rated) in the UK - (60 km^2 resolution)



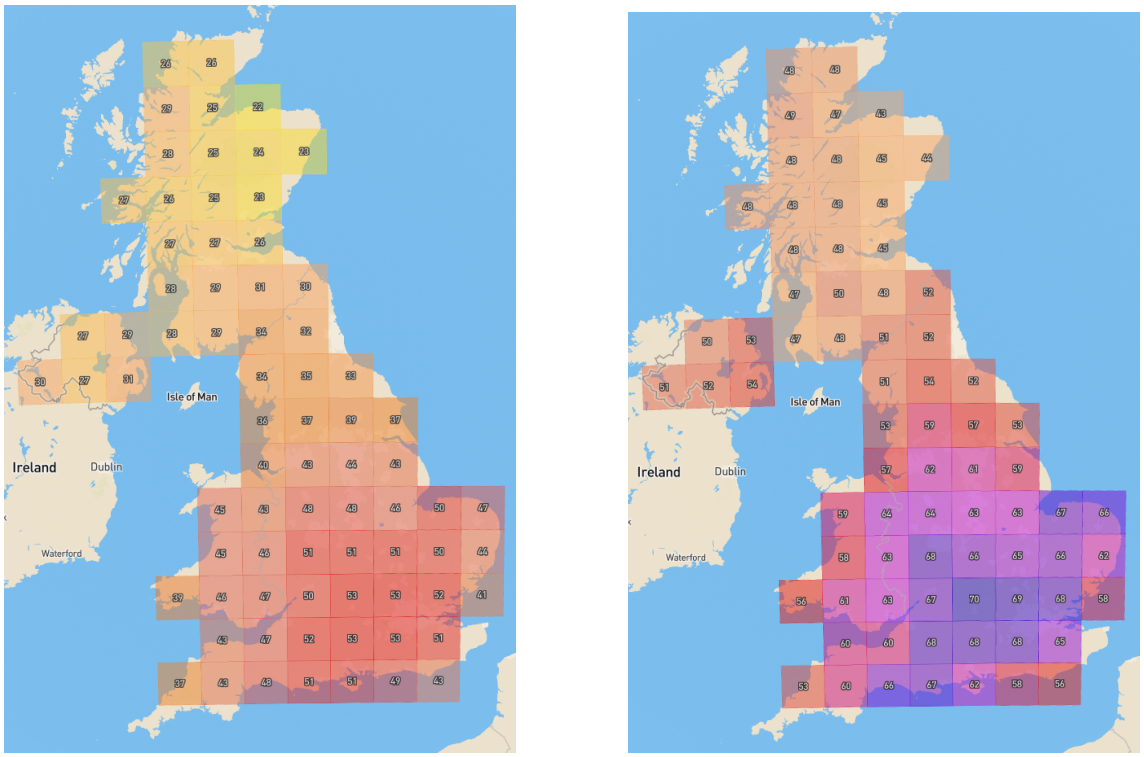
(a) 2030 – 2050 (b) 2080 – 2099

Figure 4.5: Maximum loss of load capacity for Transformers (% of rated) in the UK - (60 km^2 resolution)



(a) 2030 – 2050 (b) 2080 – 2099

Figure 4.6: Maximum loss of carrying capacity for Overhead Lines (% of rated) in the UK - (60 km^2 resolution)



(a) 2030 – 2050

(b) 2080 – 2099

Figure 4.7: Maximum increase in daily Peak Demand (% of average spring daily Peak Load) in the UK - (60 km^2 resolution)

4.4 Discussion

The chosen structure for the Grid Climate Resilience Tool allows for quick incorporation of new grid components or weather variables, provided an explicit impact function can be given to calculate each impact. There is no default built-in environment to treat impact equations needing to be solved iteratively. These would have to be solved in separate custom modules and their results included in GCRT. A desired further development for the tool is the inclusion of a Network Common Data Form (NetCDF) reader and a large-datasets handling module that would allow the loading of higher resolution datasets.

In the following the impact calculations obtained in this work are discussed. All the chosen correlations grow linearly with temperature. Therefore, an alignment of high-impact regions with high-temperature regions was expected. The magnitude of demand impact results confirms the conclusions of the NGET report, which placed temperature effects on demand second highest in its risk assessment for 2050 [20]. Results for demand are nevertheless quite uncertain, as the correlation depends strongly on the assumption made regarding average spring load for the PJM network. Additionally, most references judge that not accounting for income levels and AC penetration could lead to a significant underestimating of results [47]. To put the obtained values into perspective, an impact calculation using the correlation from [69] of 4% growth per degree above 24°C would lead to an average increase in peak load demand of 10% in 2030-2050 (max 26%) and 30% average increase (max 44%) in 2080-2099, much lower than those obtained in Section 4.3.1. This variation demonstrates the significant sensitivity of impact results to chosen correlations. In the case of demand, results could be made more precise using an empirically-built demand correlation, specific to the UK's National Grid, using historical load and temperature data.

The similar values of impact obtained for gas and nuclear generation are coherent with the fact that they are both thermoelectric generation technologies relying on cooling to increase production efficiency and, thus, have similar sensitivity mechanisms to increased ambient temperature.

The obtained impact maps show the potential perturbations of grid components if they were located in that area. However, results could be further contextualised by integrating actual or projected asset locations (or population in the case of demand). An example is provided here for nuclear capacity. The actual locations of nuclear plants [82], associated with the impact zones shown in Section 4.3.2 mean that up to 11% of nuclear capacity (1073 MWe) could be lost during high-temperature days in 2030-2050 and up to 18% (1692 MWe) could be lost in 2080-2099 (including the nuclear reactor Hickley point C into the installed capacity from 2030). A similar process could be applied to other grid components to obtain a system-wide impact.

Historically, the ratio of maximum load met in the UK to total available net capacity ranged from 70% to 79% between 2015 and 2021 [83]. This means that the remaining capacity margin was between 21% and 30%, depending on the years. Assuming that the ratios by generation apply linearly to that capacity margin, this would mean that 41% of margin capacity (covered by gas) could potentially fall by 13% on average in 2030-2050 (resp. 18% in 2080-2099) and 17% of margin capacity (covered by nuclear power) could potentially fall by 13% on average in 2030-3050 (resp. 18% in 2080-2099). Under these assumptions, the generation reserve margin could fall by 9% over 2030-2050 and 12% over 2080-2099 on the highest-temperature days. These values indicate that some capacity margin would remain even on the highest-impact days if accounting for generation reduction only. Accounting for increased demand and reduced transmission capacities could, however, mean that certain UK areas would not be able to cover demand even in the 2030-2050 period if no adaptation measures are taken. In order to keep a capacity margin above 20%, serious adaptation of generation capacity would be needed, especially in the southeast of the country, where high-temperature impacts will be the highest.

4.5 Conclusion and outlooks

The completion of this work demonstrates that it is possible to develop an adaptable program calculating climate impacts on electricity grid components and associate it with an interactive visualisation tool that could be used by the wider public. The encouraging achieved flexibility demonstrates that the GCRT provides a robust basis for climate risk visualisation on electricity systems and infrastructure assets in general. This work delivers a successful proof-of-concept for analysing potential impacts on UK grid components, focusing on ambient temperature effects.

The implementation of the program for the case of ambient temperature shows that high-temperature extremes projected over the UK in a high-emissions (RCP8.5) scenario will considerably impact electricity grid components and peak demand. Impacts will be higher in the country's southeast, where temperatures will be the highest. However, even the lowest temperature regions (Scotland and Northern Ireland) will lose significant generation and transmission capacities, with a simultaneous increase in daily peak demand. Impacts on all grid components are more severe in the 2080-2099 period than in the 2030-2050 period.

Taken in isolation, the projected impacts on generation could be absorbed by existing reserve capacity margins in the UK. However, the simultaneous loss of transmission capacity and increased peak demand could challenge network balancing. High-impact days may push operators to rely on energy imports (which may not be available if similar impacts affect neighbouring countries and their power generation) or to trigger blackouts to protect the grid during the highest-temperature days.

Opportunities for further developments based on the results of this work include:

- carrying a similar analysis on a broader geographical scale (European or global scales);
- introducing additional grid components (like wind and hydroelectric generation) or weather variables (like water availability) to the study;
- iterating the analysis at a greater spatial resolution to capture local effects;
- calculating impact in terms of the length and frequency of "hot days" as well as their intensity;
- incorporating actual asset location and population levels to produce a system-wide impact analysis.

Such extensions could bring the developed tool up to its full potential so that it can be optimally exploited and contribute to better prepare society for the unavoidable weather pressures brought by climate change.

Coming back to the comment by Dr Hebner after the Texas Power Crisis, which partly inspired this work: *"The real outstanding question nationally is, given climate change, how much more extreme weather should we be protecting against".* it was the author's wish and ambition to provide an answer that contributes to the increase of capacity to cope with extreme-weather impacts, minimising their detrimental effects on human well-being.

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